Toward Building ***Deletion Compliant Data Systems***

Subhadeep Sarkar

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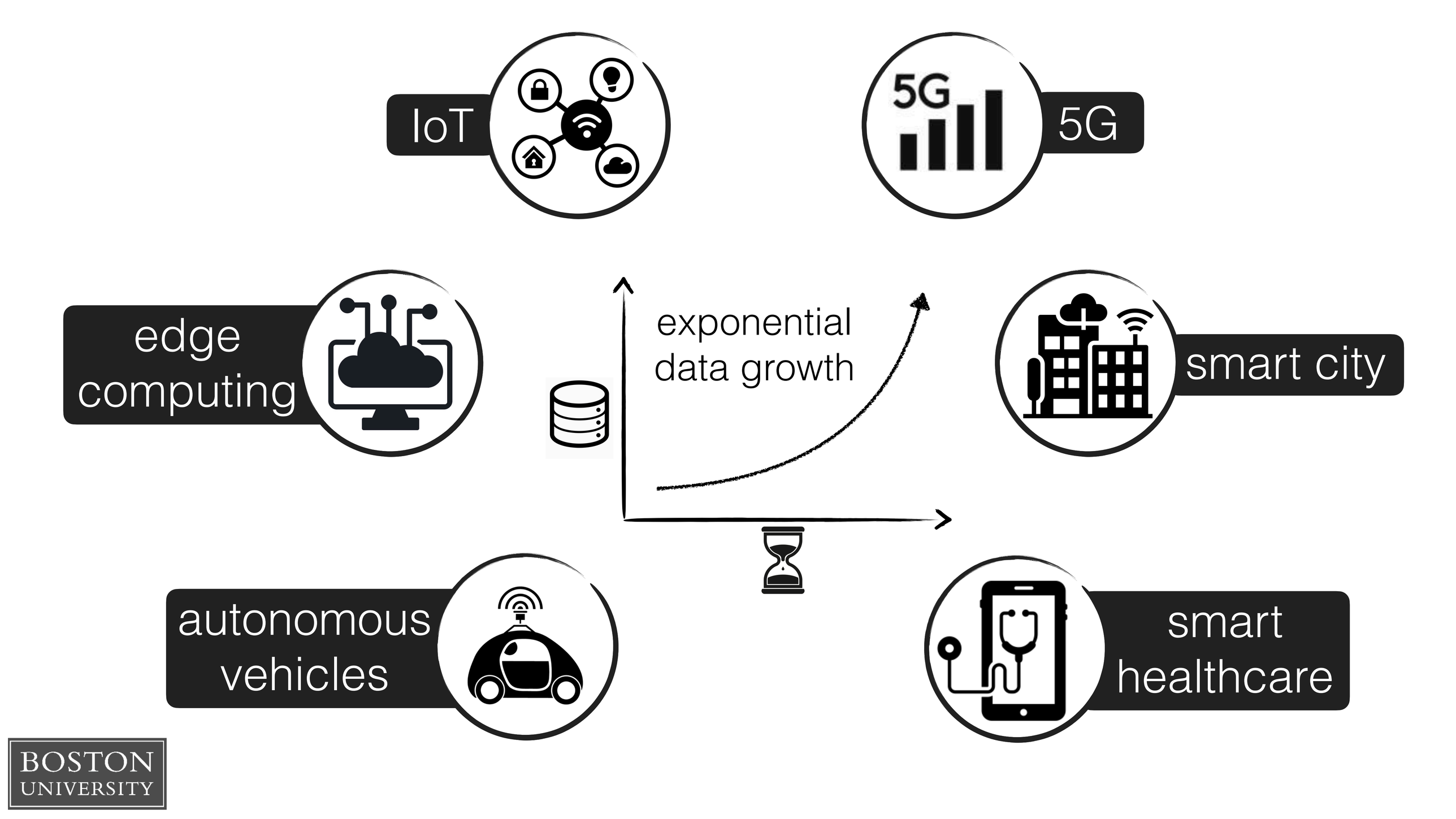
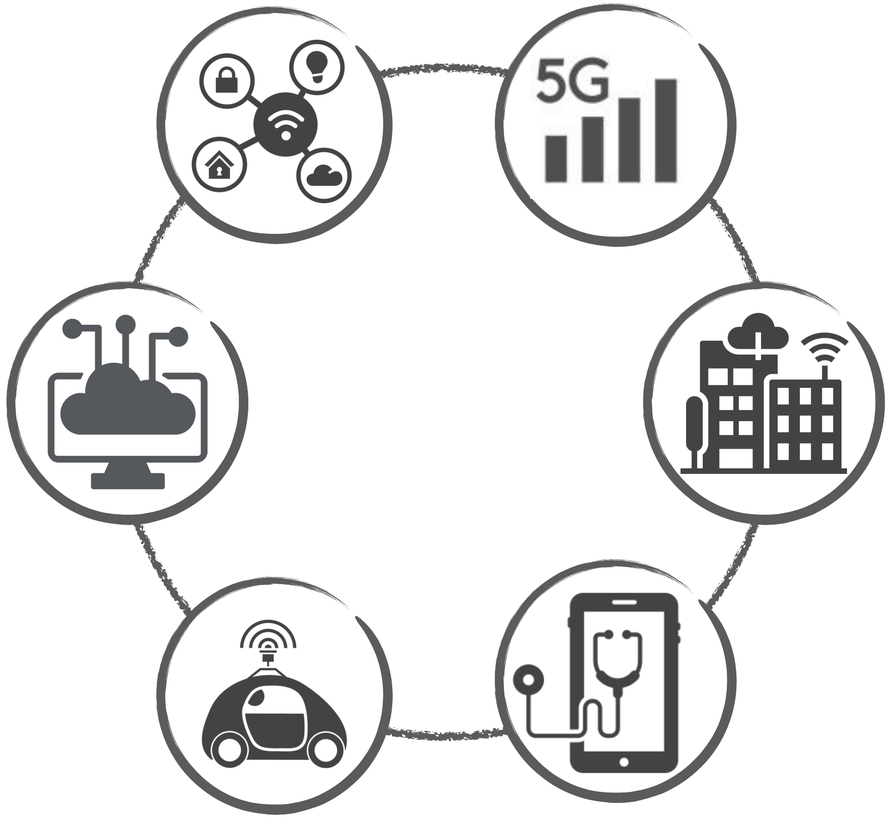
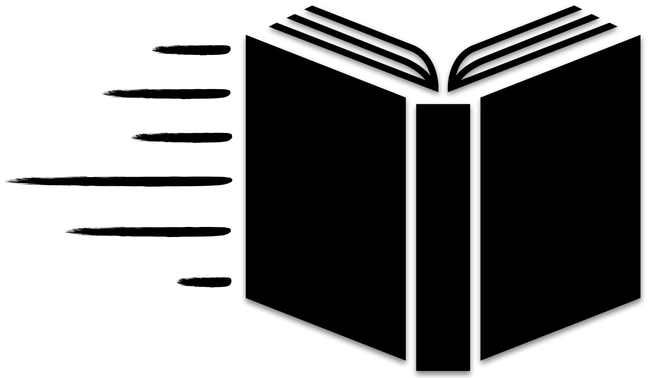
DiSC lab


IoT
5G
5G
edge computing
exponential data growth
smart city
autonomous vehicles
smart healthcare



fast writes



fast reads

Out-of-place paradigm

Relational & Array-based

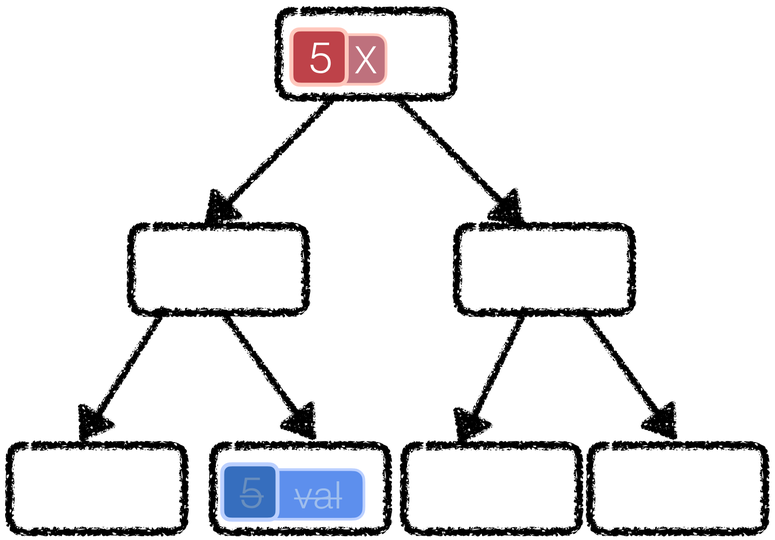
NoSQL

Bigtable

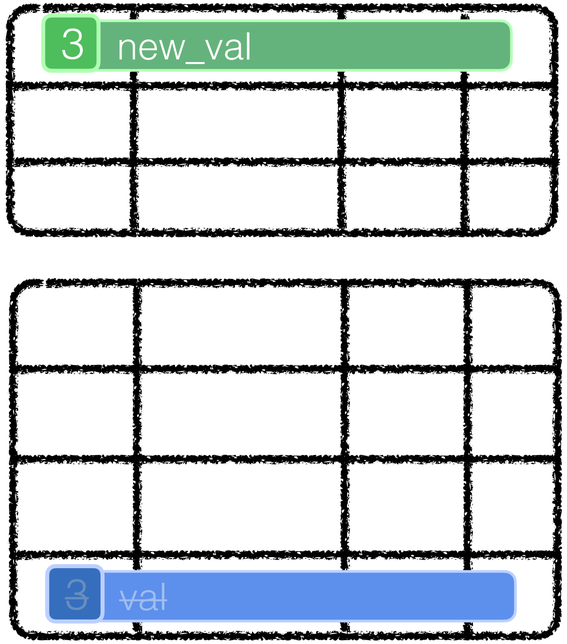
Out-of-place paradigm

Out-of-place updates/deletes

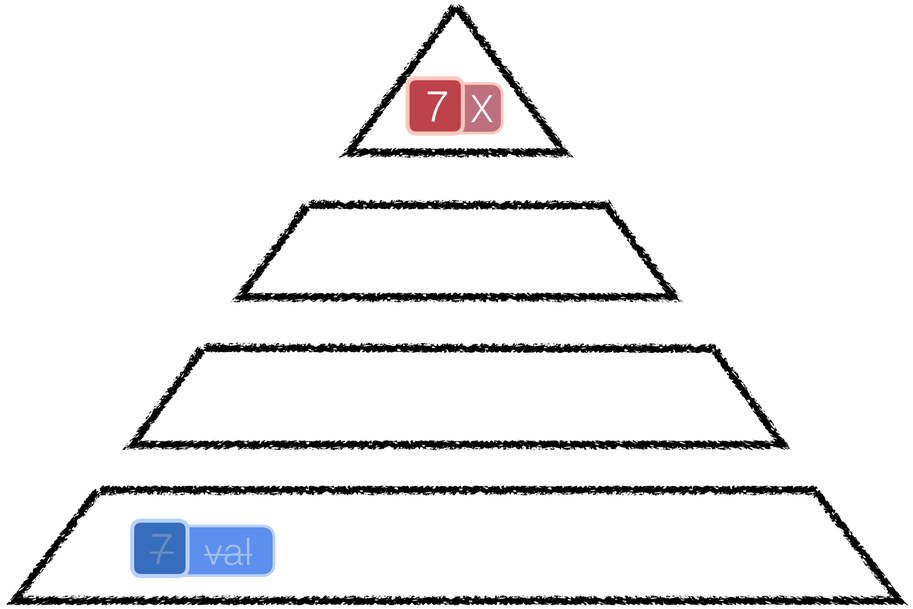
Merge updates/deletes lazily



BetrFS



VERTICA



RocksDB

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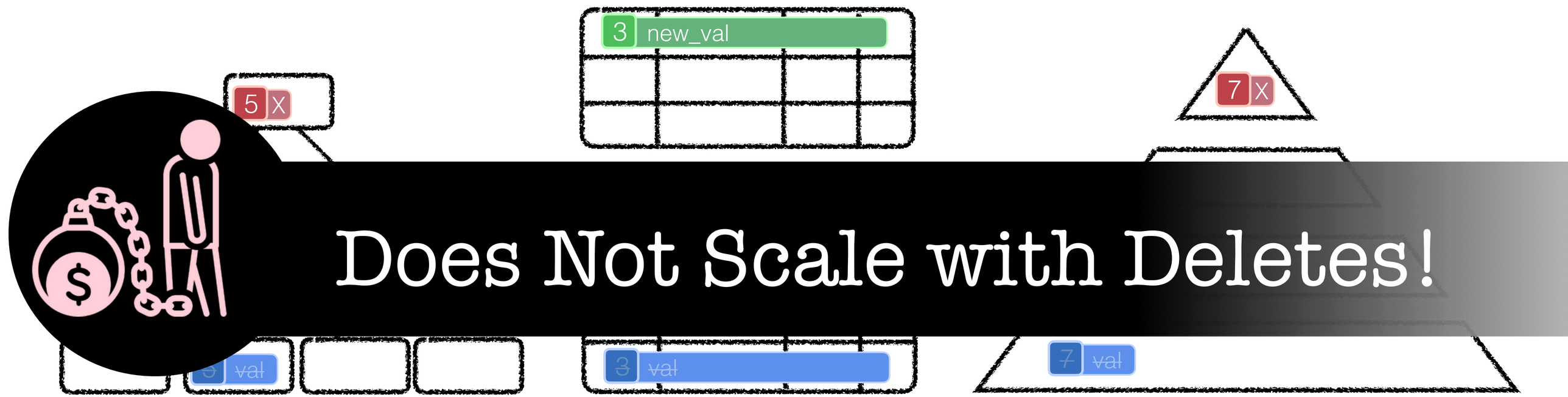
Out-of-place paradigm

Out-of-place updates/deletes

Merge updates/deletes lazily



Does Not Scale with Deletes!

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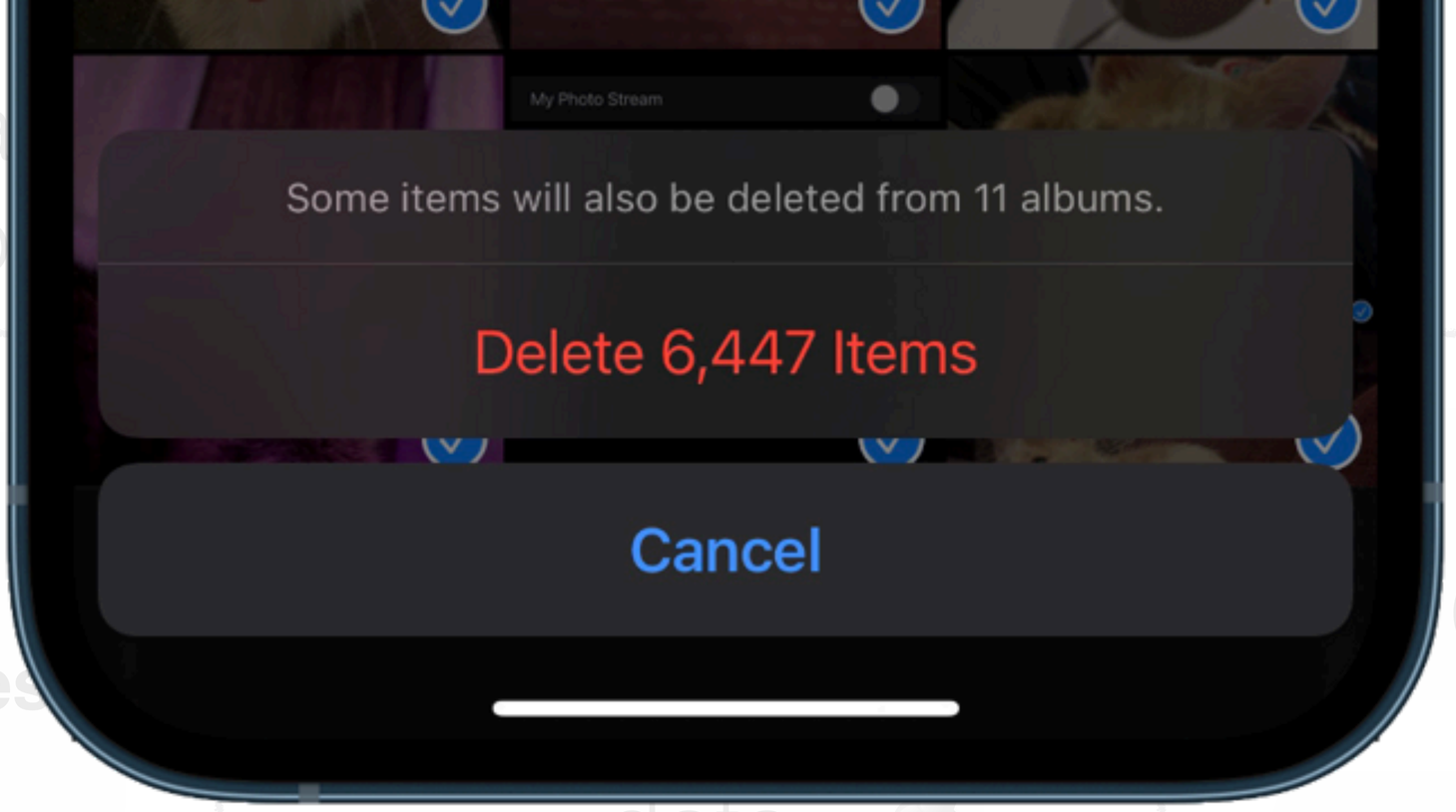
BetrFS

VERTICA

RocksDB



Large-sca
productio

Privacy

ZippyDB

25.2M deletes
day

GDPR
(EU, UK)

data
migration

CCPA
(California)

UP2X

100M merge via
deletes / day

periodic
cleanup

VCDPA
(Virginia)

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Large-scale production

ZippyDB

25.2M deletes / day

UP2X

100M merge via deletes / day

Internal DB ops

table drop

data migration

periodic cleanup

Privacy

GDPR (EU, UK)

CCPA (California)

VCDPA (Virginia)

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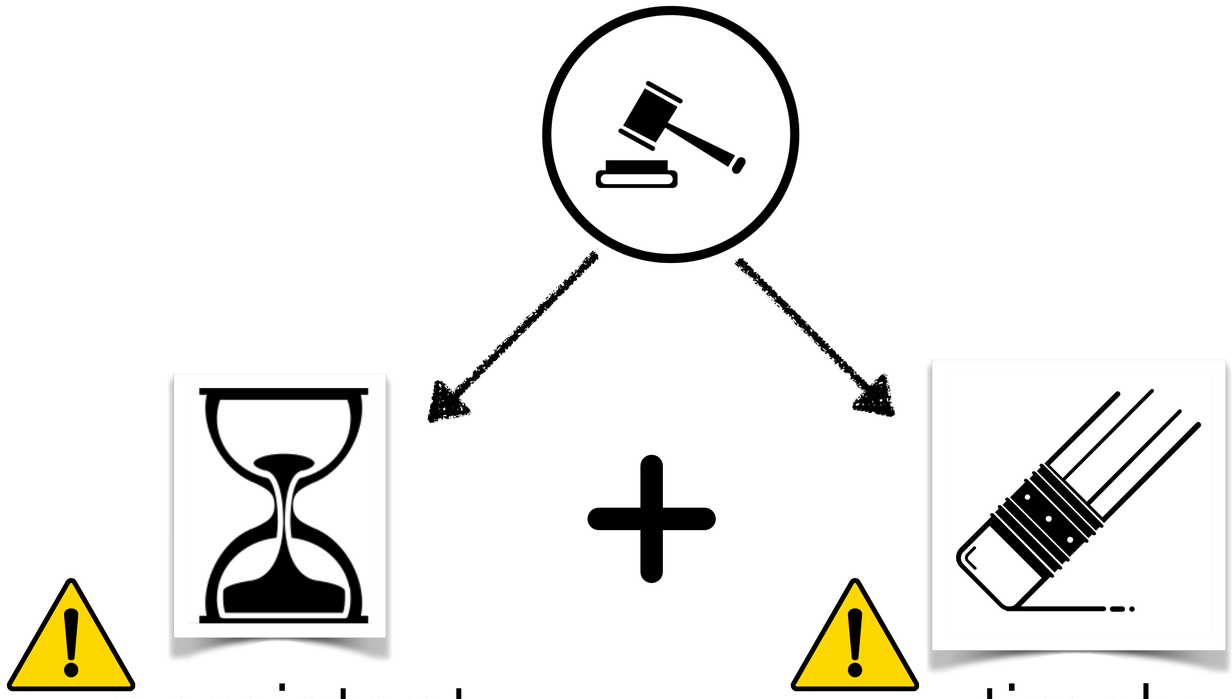
GDPR
(EU, UK)

CCPA
(California)

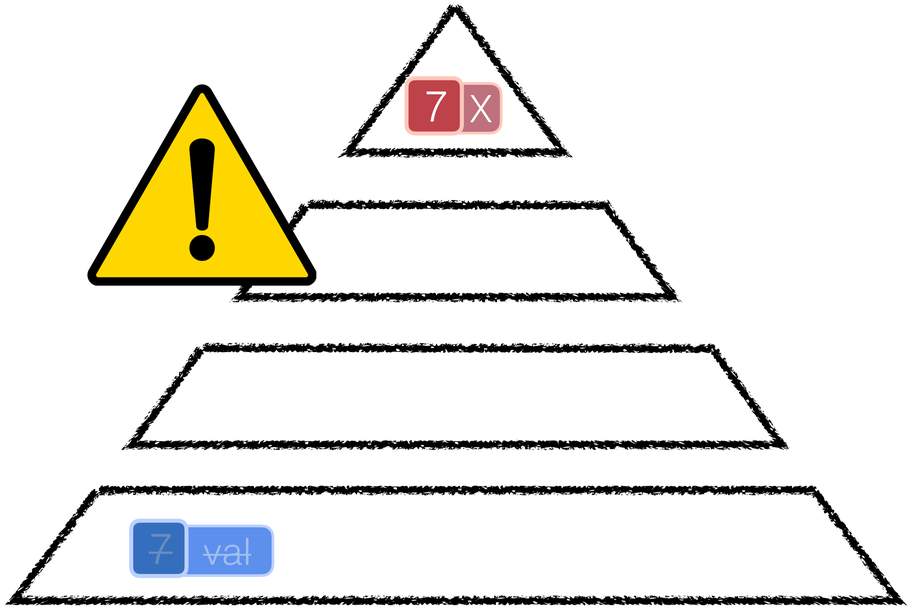
VCPDA
(Virginia)

MANDATES



persistent
deletes

timely
deletes



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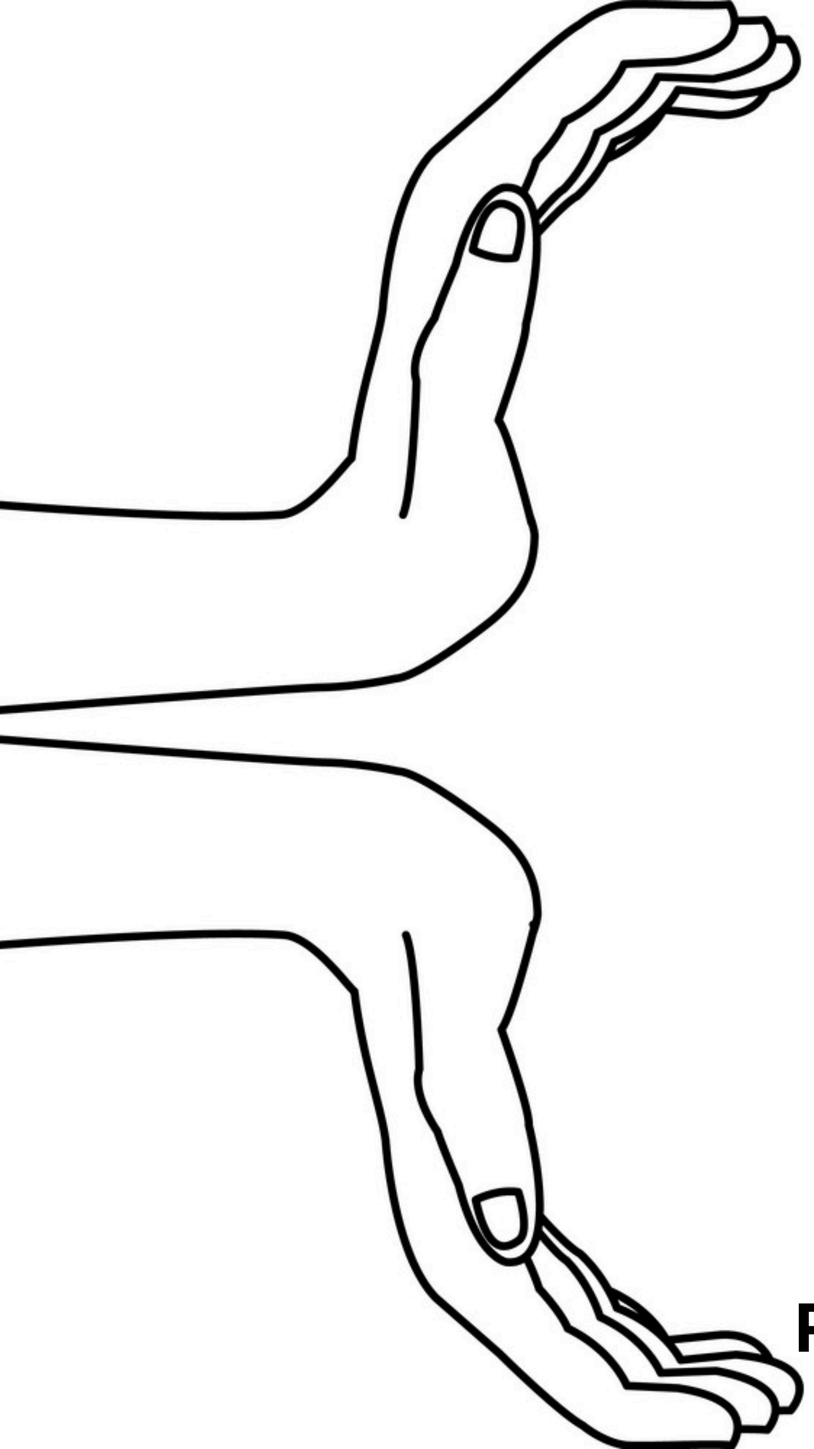
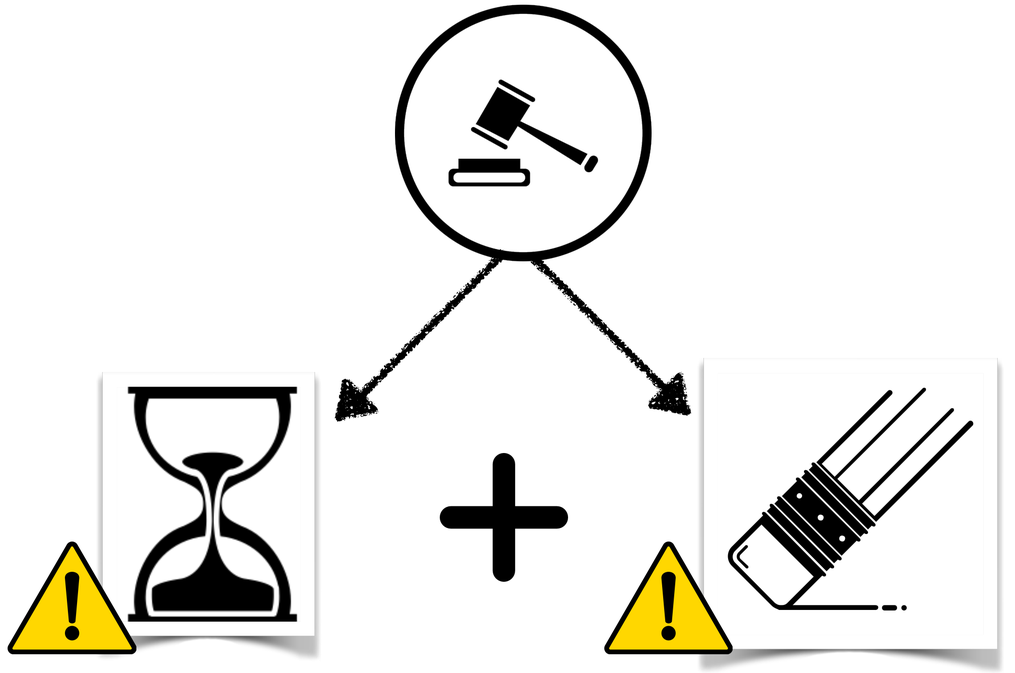




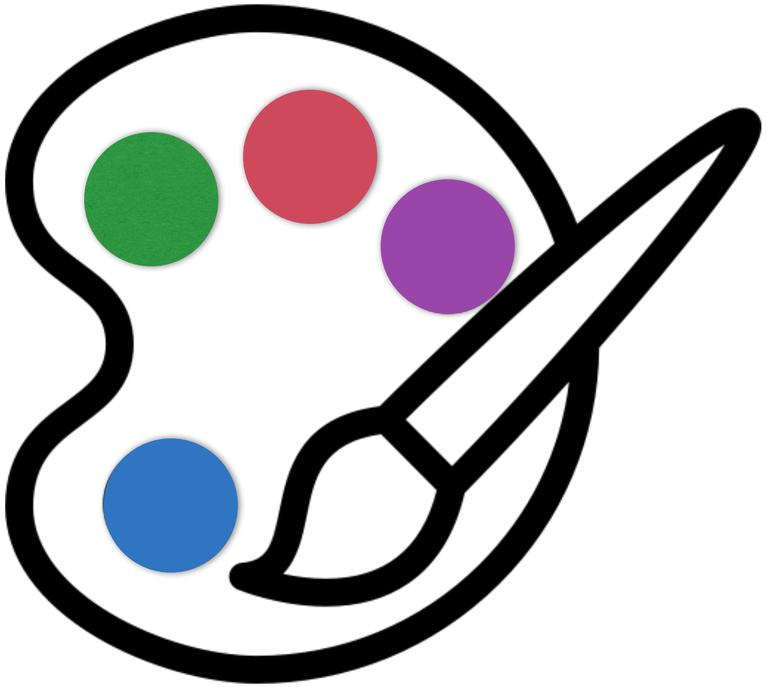
persistent deletes

timely deletes

PRIVACY **PERFORMANCE** **COST**

Our **Vision**

PRIVACY-THROUGH-DELETION



- Navigate deletion requirements
- Translate the requirements
- Realize efficient deletes
- Scale with data

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Regulations

Right to be forgotten

Right to delete

Deletion right

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Regulations

Right to be forgotten

Right to delete

Deletion right

Extracting Requirements

on-demand

retention-based

BOSTON UNIVERSITY

Regulations

Right to be forgotten

Right to delete

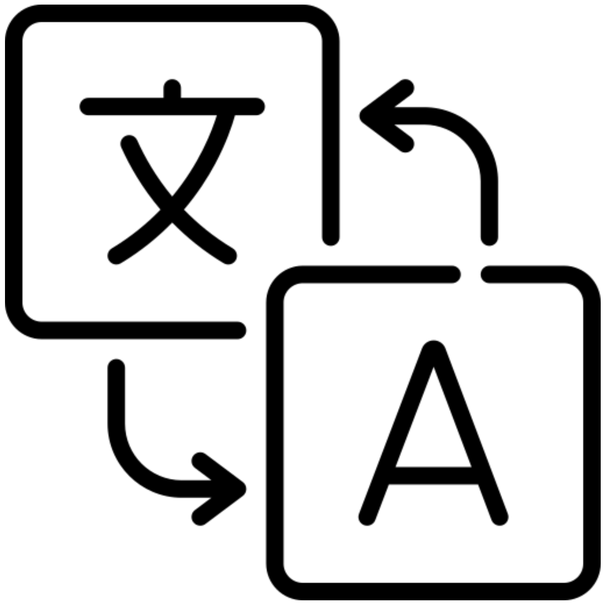
Deletion right

Extracting Requirements

on-demand

retention-based

Query Language Support



requirement translation

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SQL-support for deletes

```
CREATE TABLE R (column1 type1, column2 type2, ...)
  WITH RET_DUR
    {ARBITRARY|FIXED (t1 <ret1>, t1 <ret1>, ...)}
  WITH DPT
    {ARBITRARY|FIXED (d1 <dpt1>, d1 <dpt1>, ...)};
```

```
INSERT INTO R (val1, val2, ...)
  WITH RET_DUR {<t>|t<i>};
```

```
DELETE FROM R
  WHERE (...)
    WITH DPT {<d>|d<i>};
```

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Regulations

Right to be forgotten

Right to delete

Deletion right

Extracting Requirements

on-demand

retention-based

Query Language Support

```
CREATE TABLE R (...)
   WITH RET_DUR
   {ARBITRARY | FIXED(...)}
   WITH DPT
   {ARBITRARY | FIXED(...)};
```

```
INSERT INTO R (...)
   WITH RET_DUR {<t>|t<i>};
```

```
DELETE FROM R
   WHERE (...)
      WITH DPT {<d>|d<i>};
```

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Regulations

Right to be forgotten

Right to delete

Deletion right

Extracting Requirements

on-demand

retention-based

Query Language Support

```
CREATE TABLE R (...)
   WITH RET_DUR
   {ARBITRARY | FIXED(...)}
   WITH DPT
   {ARBITRARY | FIXED(...)};
```

```
INSERT INTO R (...)
   WITH RET_DUR {<t>|t<i>};
```

```
DELETE FROM R
   WHERE (...)
      WITH DPT {<d>|d<i>};
```

System Support

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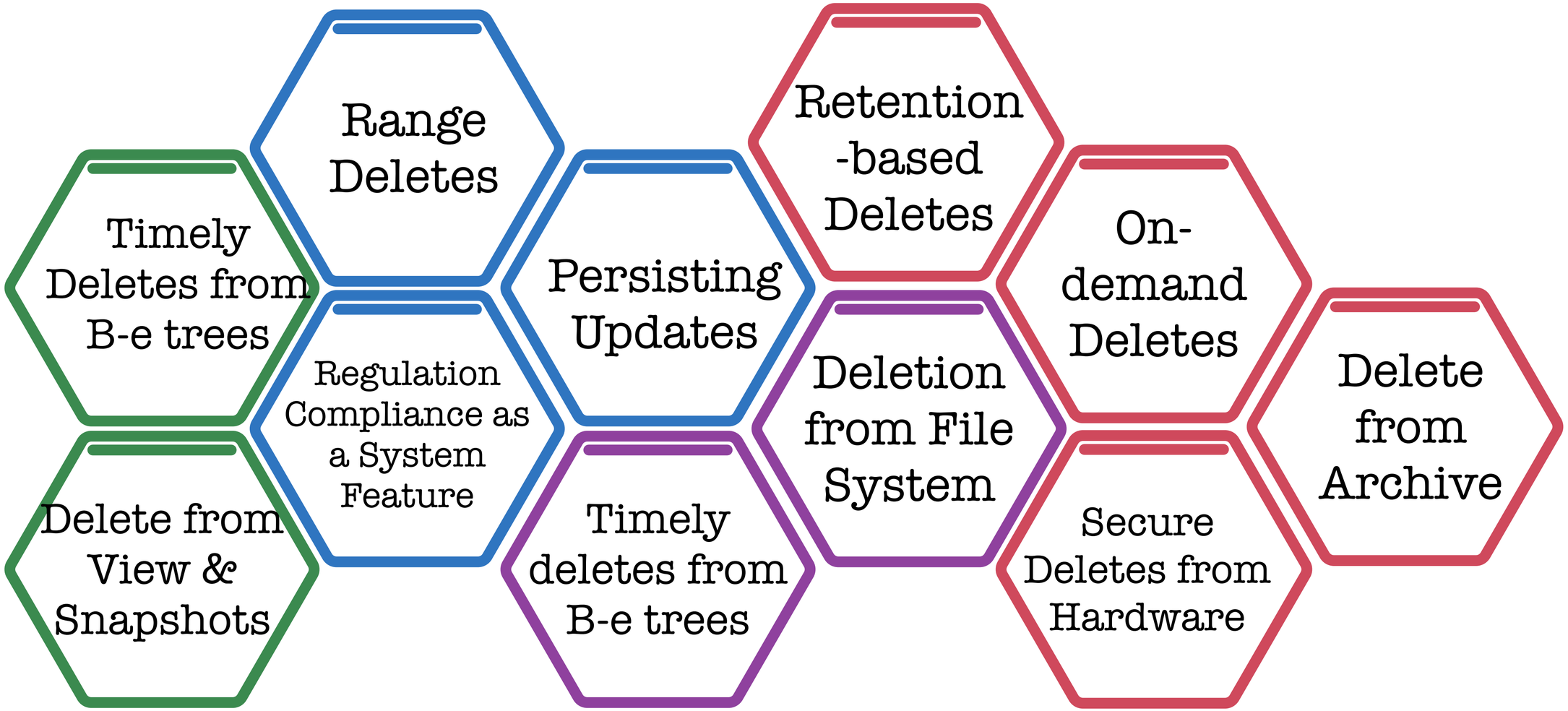

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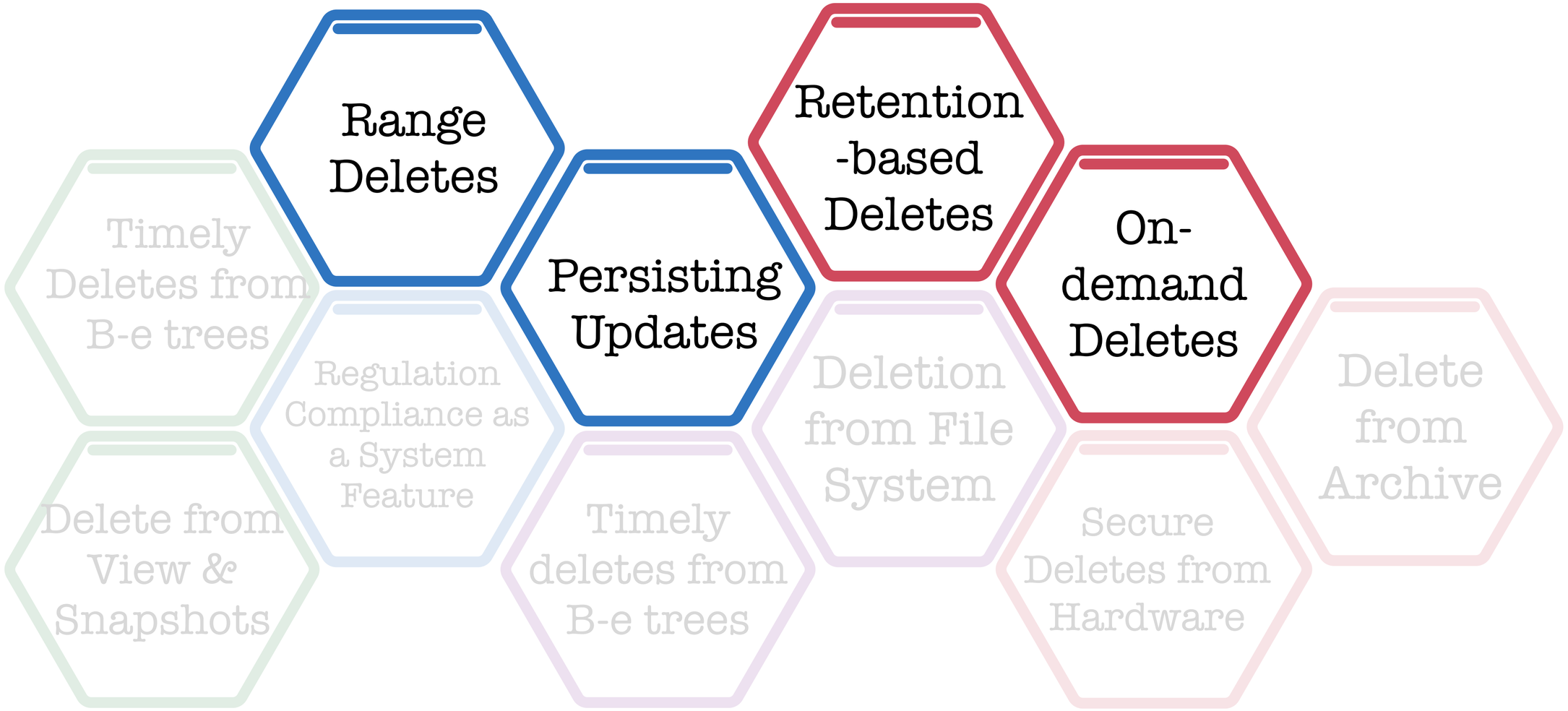
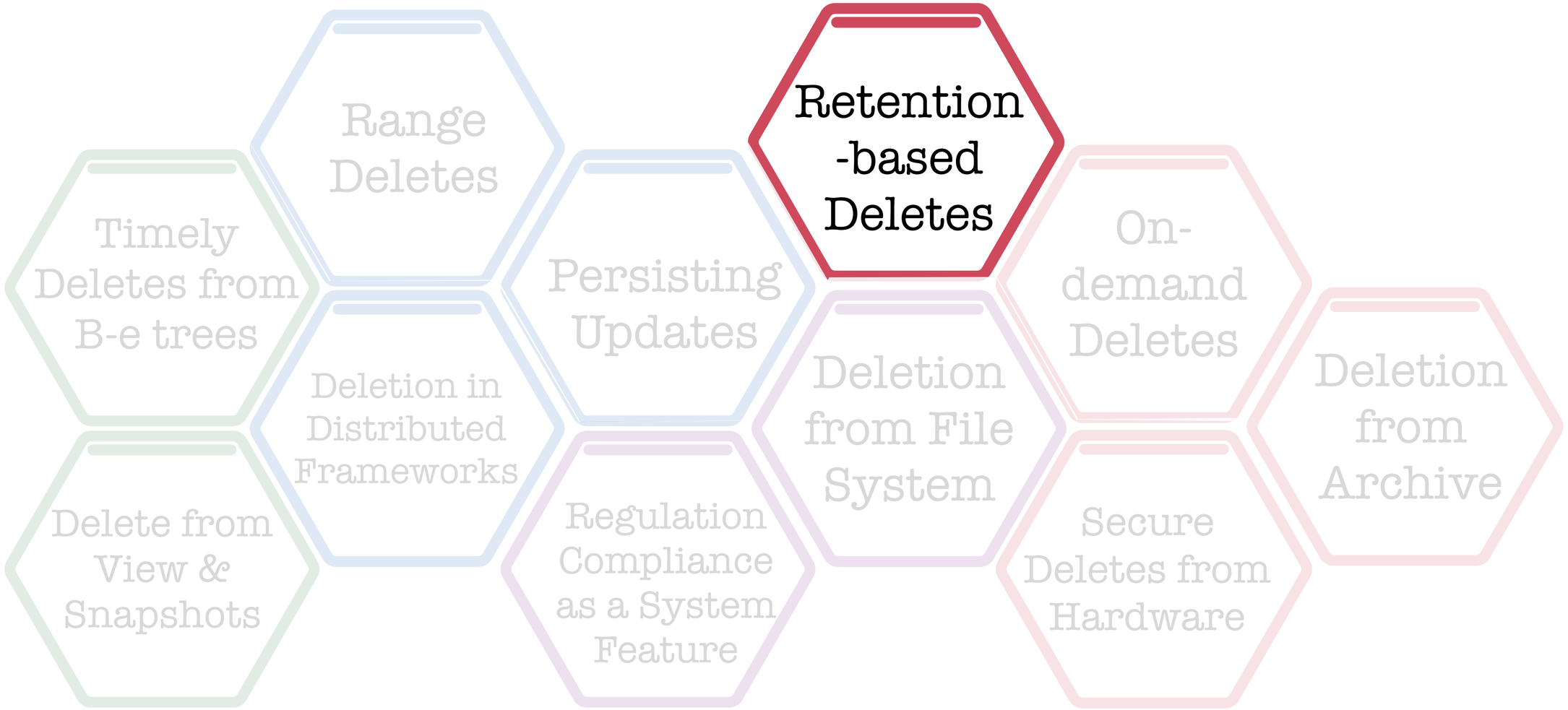

Range Deletes
Retention -based Deletes
Persisting Updates
On-demand Deletes
Timely Deletes from B-e trees
Regulation Compliance as a System Feature
Deletion from File System
Delete from Archive
Delete from View & Snapshots
Timely deletes from B-e trees
Secure Deletes from Hardware



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Log-**S**tructured **M**erge-tree

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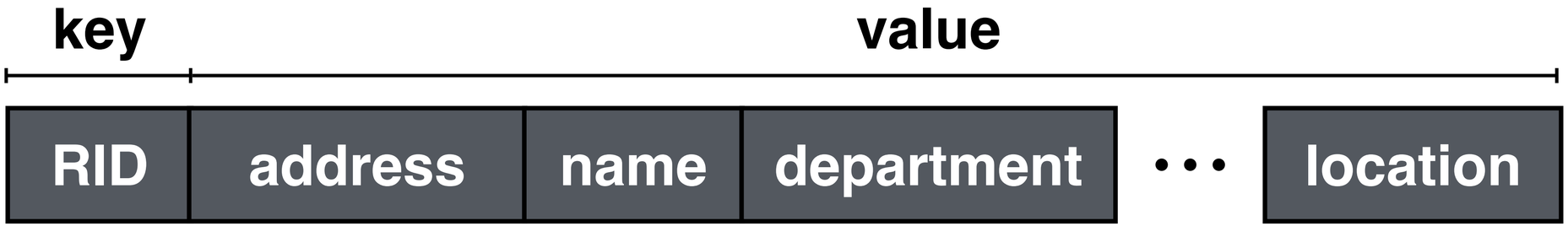

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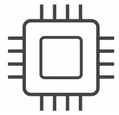
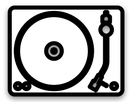


buffer

2	6	1	4



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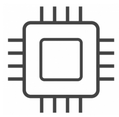
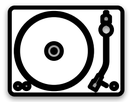


buffer

1	2	4	6



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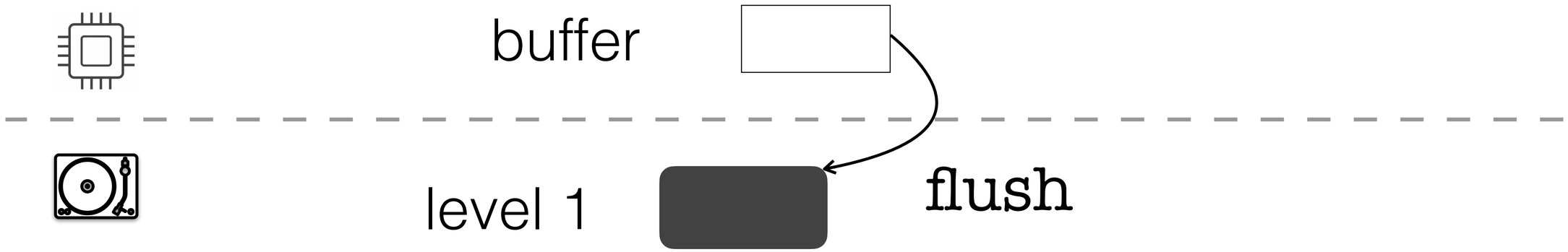

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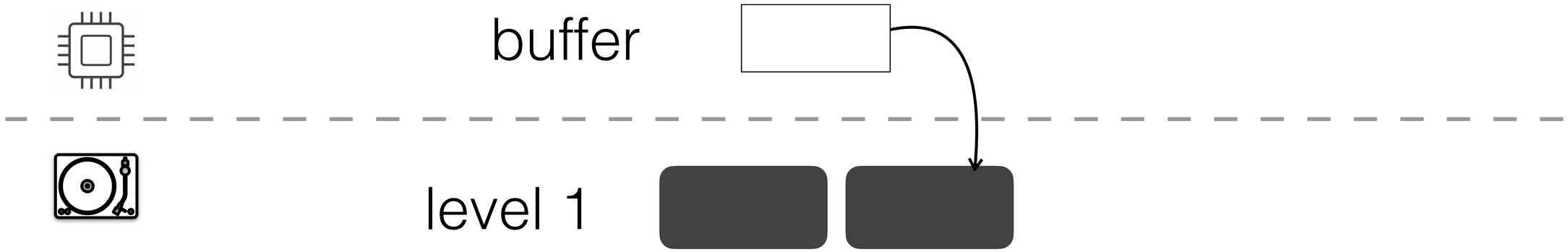

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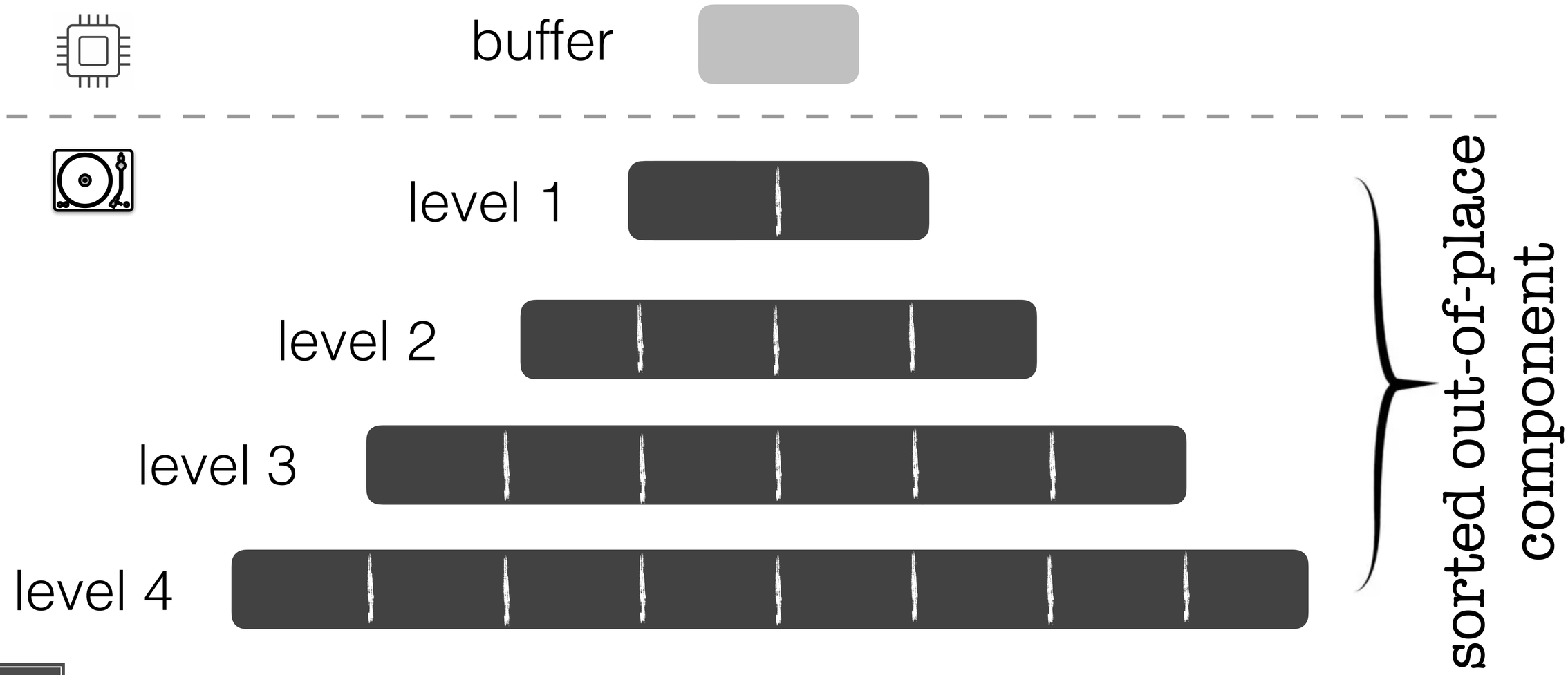
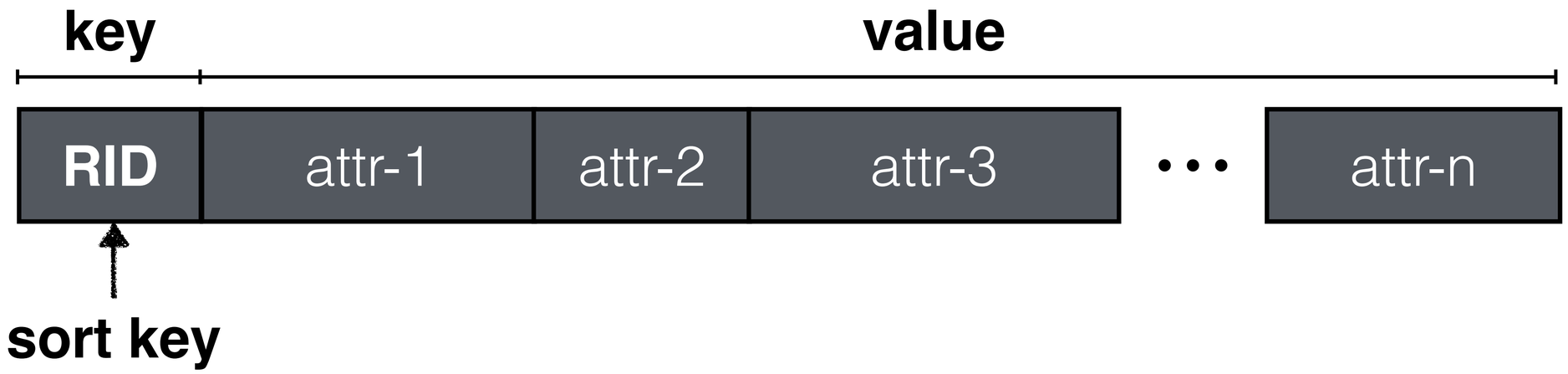

buffer
level 1
level 2
level 3
level 4
sorted out-of-place component

Realizing **Retention-Based Deletes**

delete all entries older than: TS_x



sort key = delete key

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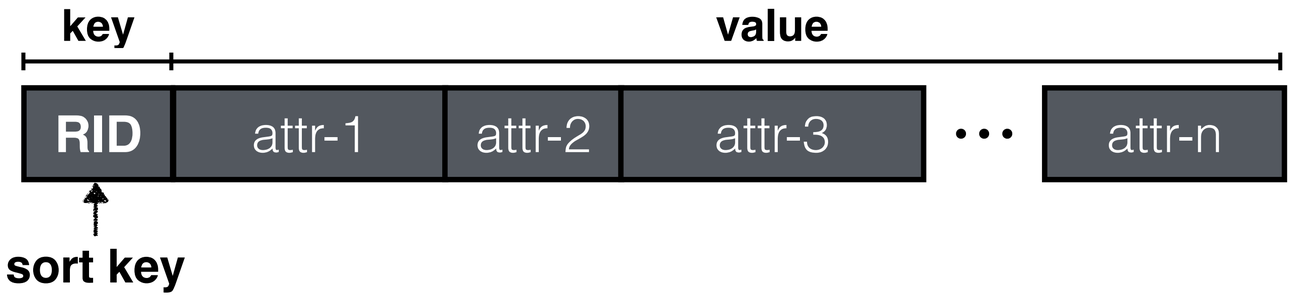
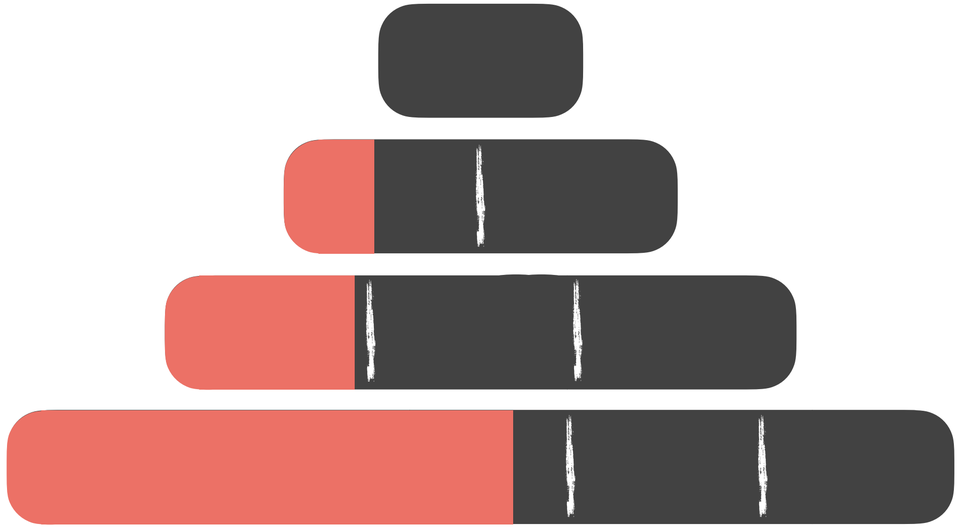
Realizing **Retention-Based Deletes**

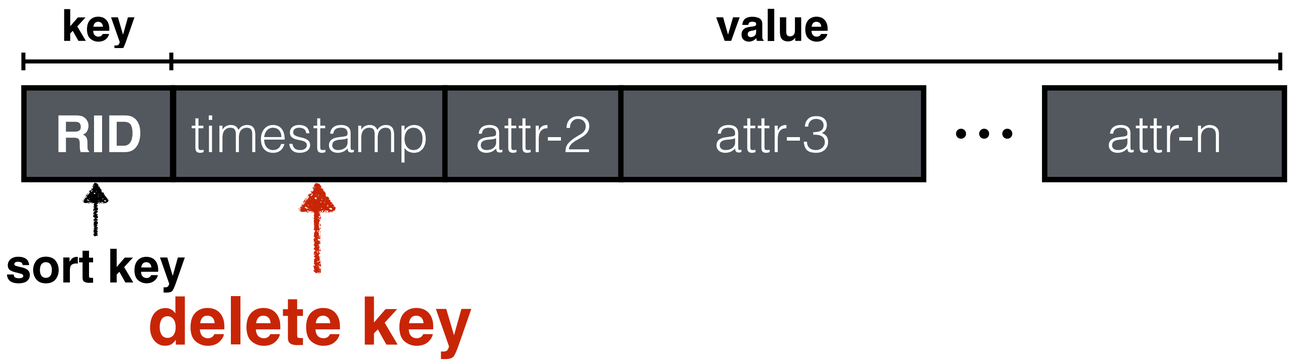
delete all entries older than: TS_x



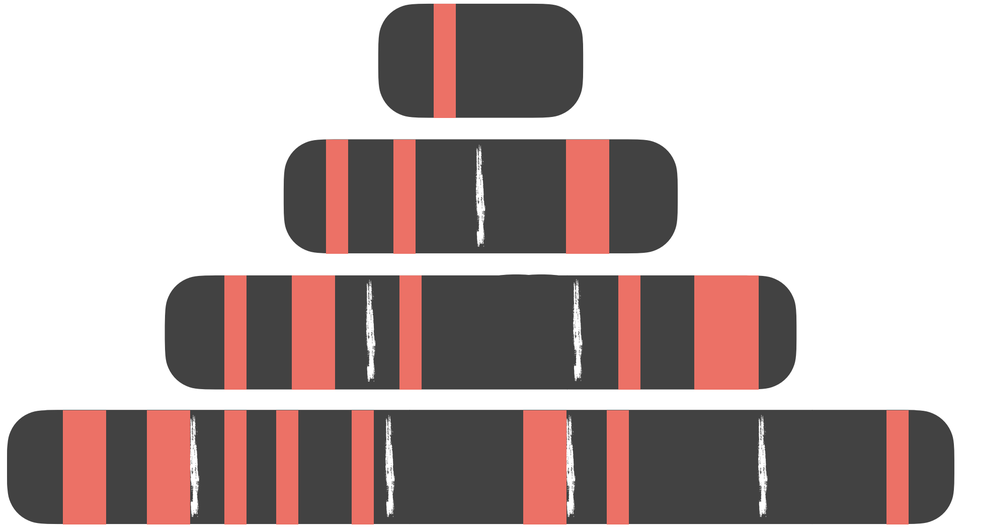


Realizing **Retention-Based Deletes**

delete all entries older than: TS_x



sort key ≠ delete key



scattered occurrences

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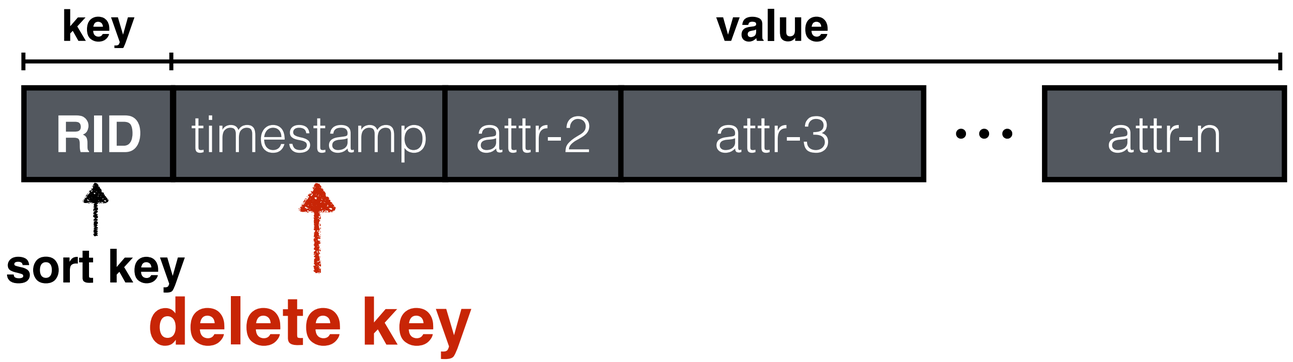
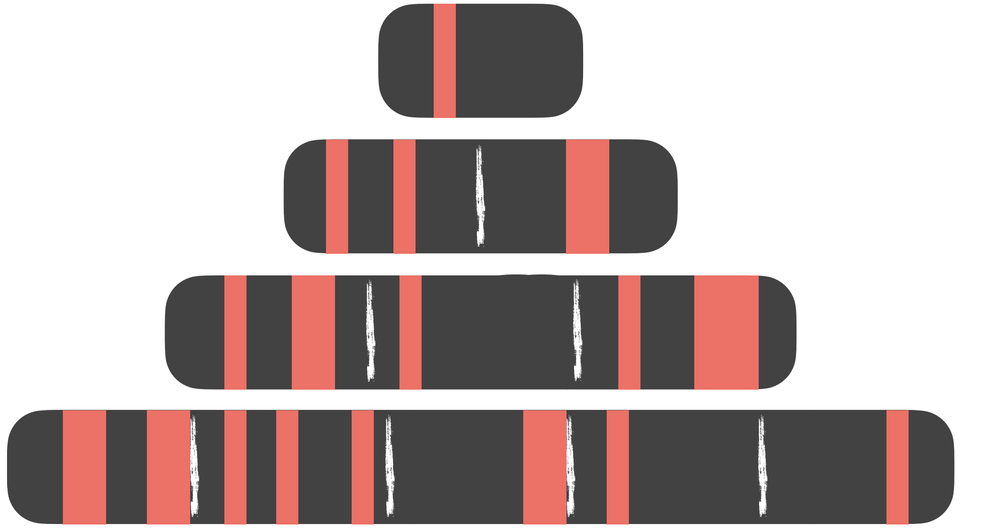
Realizing **Retention-Based Deletes**

delete all entries older than: TS_x



sort key ≠ delete key



scattered occurrences

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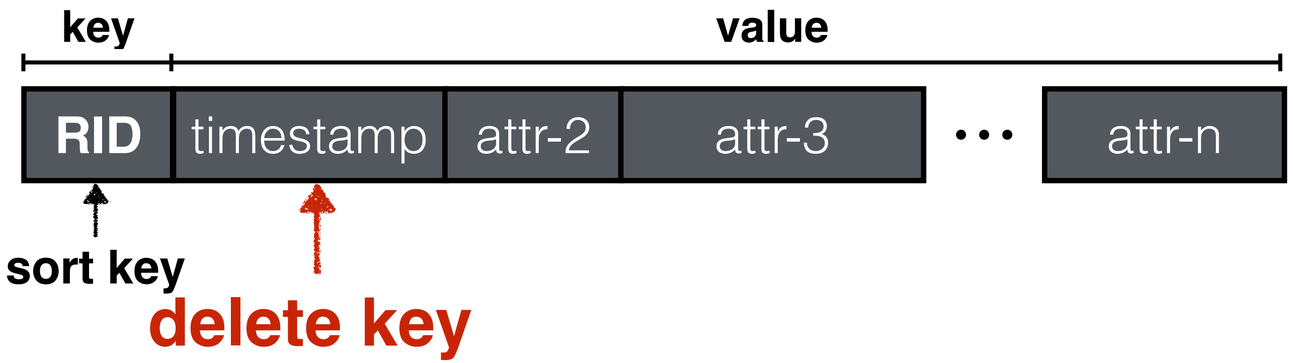
Realizing **Retention-Based Deletes**

delete all entries older than: TS_x



sort key ≠ delete key

latency spikes ✗

superfluous I/Os ✗

Realizing **Retention-Based Deletes**

delete all entries older than <= 65_D

SST file

S_{min}=1 :: S_{max}=99
D_{min}=1_D :: D_{max}=90_D

page 1: S_{min}=1 :: S_{max}=24, D_{min}=3_D :: D_{max}=80_D

page 2: S_{min}=29 :: S_{max}=60, D_{min}=9_D :: D_{max}=90_D

page 3: S_{min}=61 :: S_{max}=79, D_{min}=1_D :: D_{max}=89_D

page 4: S_{min}=80 :: S_{max}=99, D_{min}=7_D :: D_{max}=85_D

page 1

1	4	9	14	15	19	20	24
34_D	69_D	3_D	79_D	8_D	80_D	23_D	24_D

page 2

29	32	33	40	44	52	56	60
88_D	90_D	28_D	74_D	9_D	76_D	81_D	64_D

page 3

61	63	67	71	72	73	78	79
75_D	82_D	1_D	67_D	77_D	89_D	65_D	12_D

page 4

80	84	86	87	91	94	95	99
70_D	41_D	62_D	7_D	25_D	85_D	59_D	19_D

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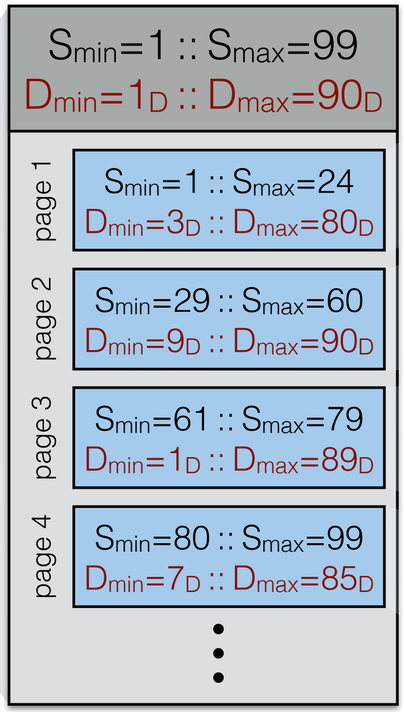
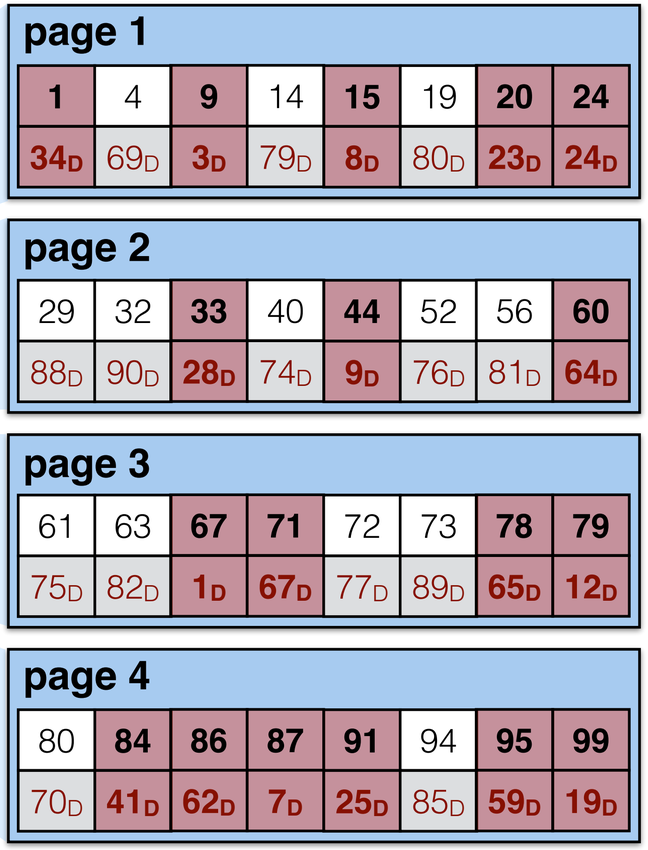
Realizing **Retention-Based Deletes**

delete all entries older than <= 65_D

SST file





1 I/O

1 I/O

1 I/O

1 I/O

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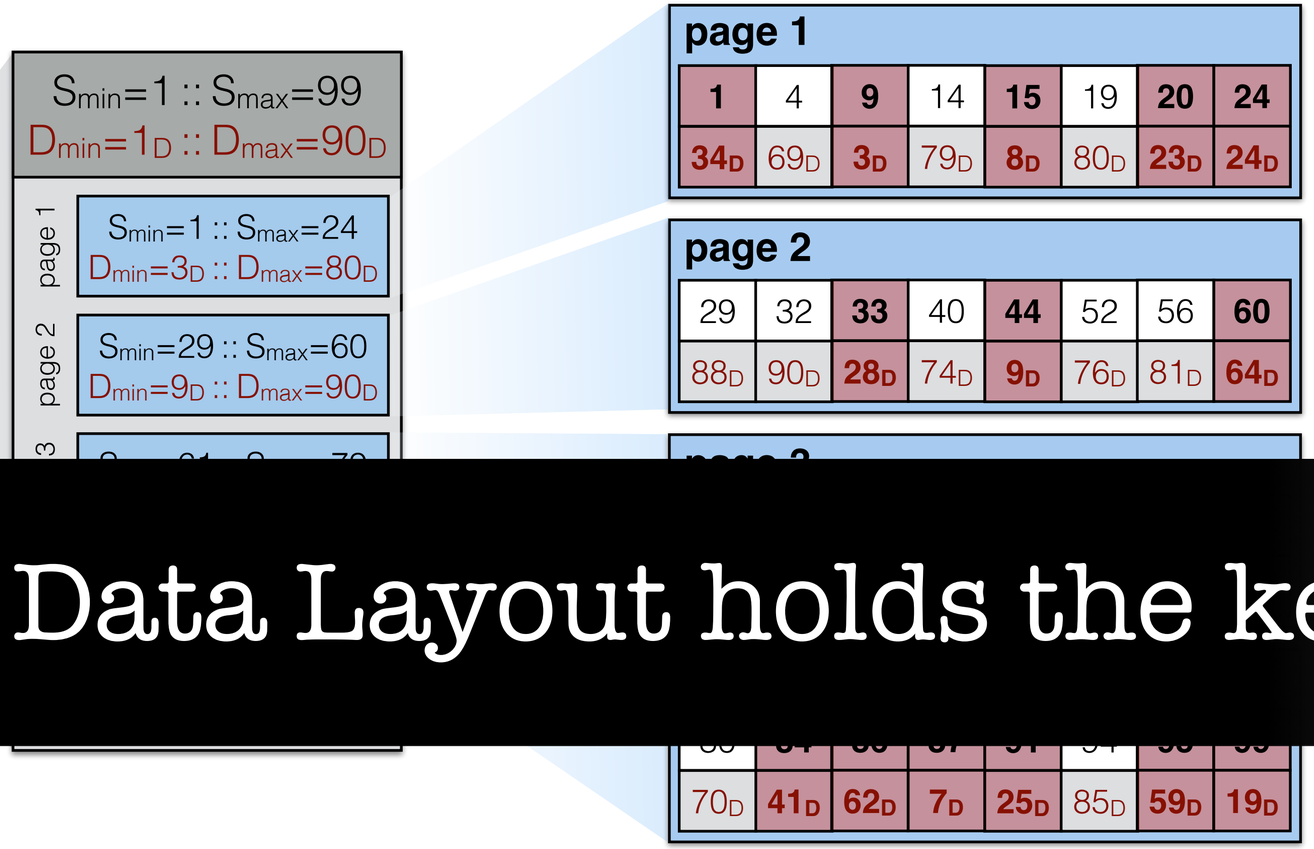
Realizing **Retention-Based Deletes**

delete all entries older than <= 65_D



Data Layout holds the key!

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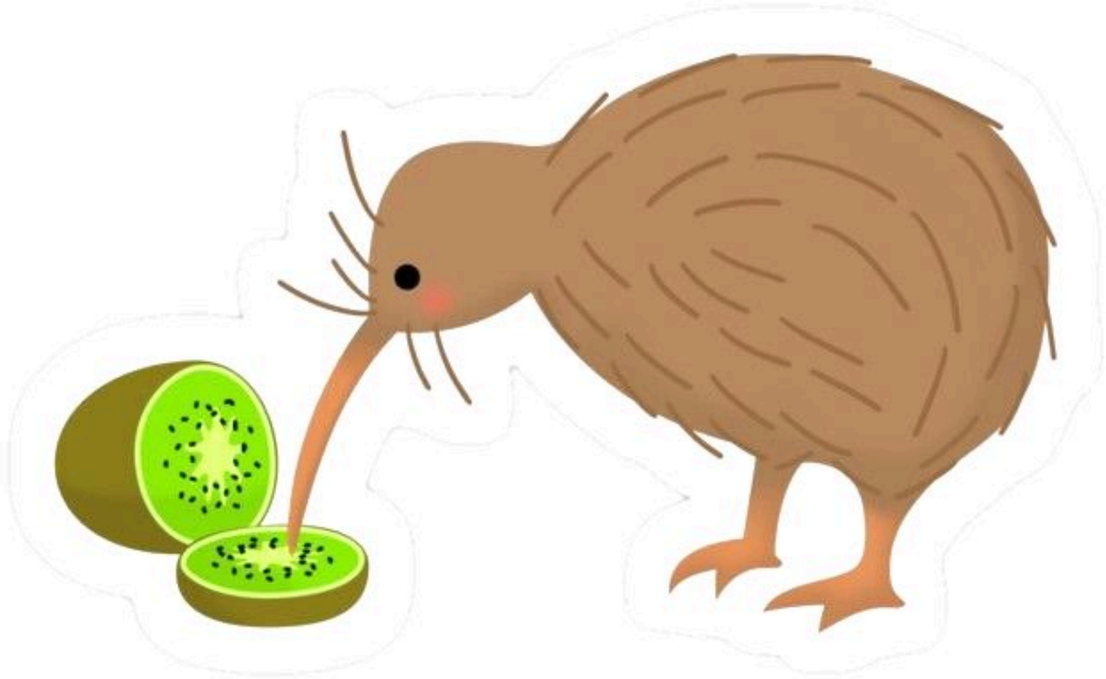
Realizing **Retention-Based Deletes**



KiWi

Key Weaving storage layout

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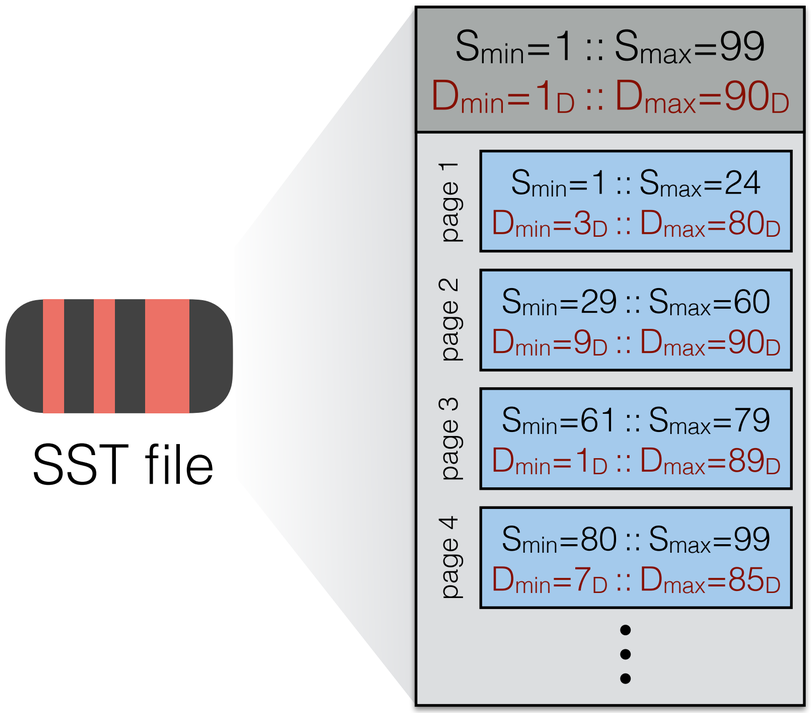
Key Weaving storage layout

delete all entries older than <= 65_D



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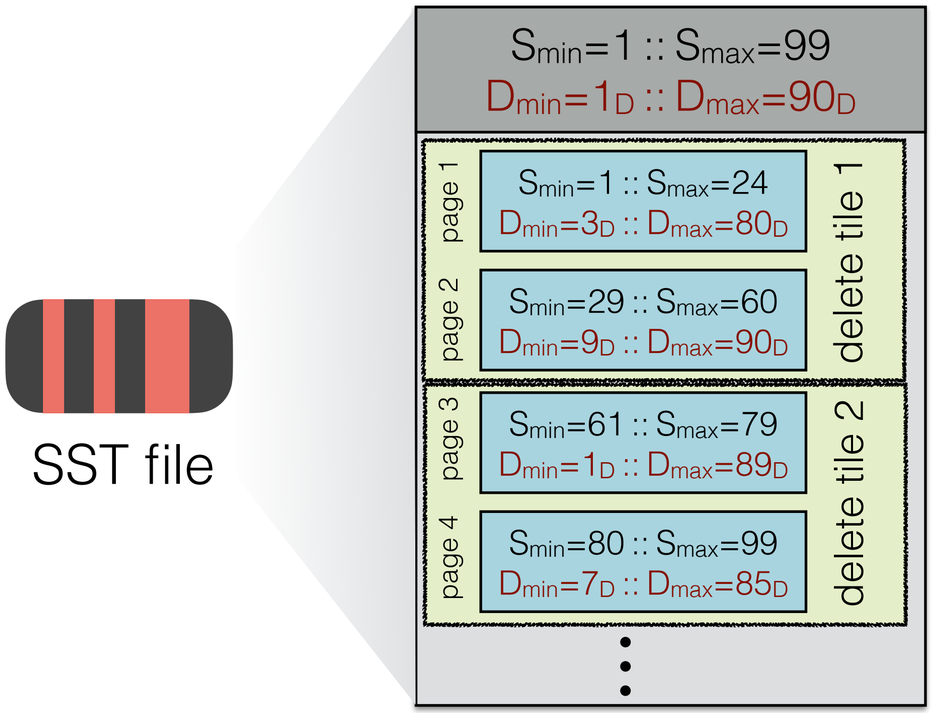
Key Weaving storage layout

delete all entries older than <= 65_D



partitioned on S

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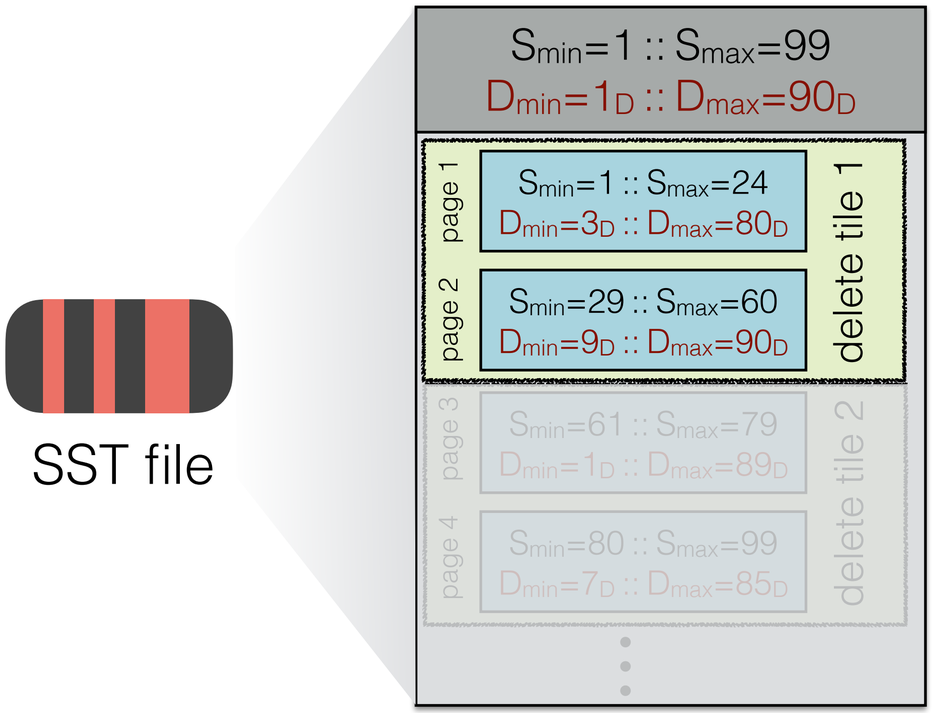
Key Weaving storage layout

delete all entries older than <= 65_D



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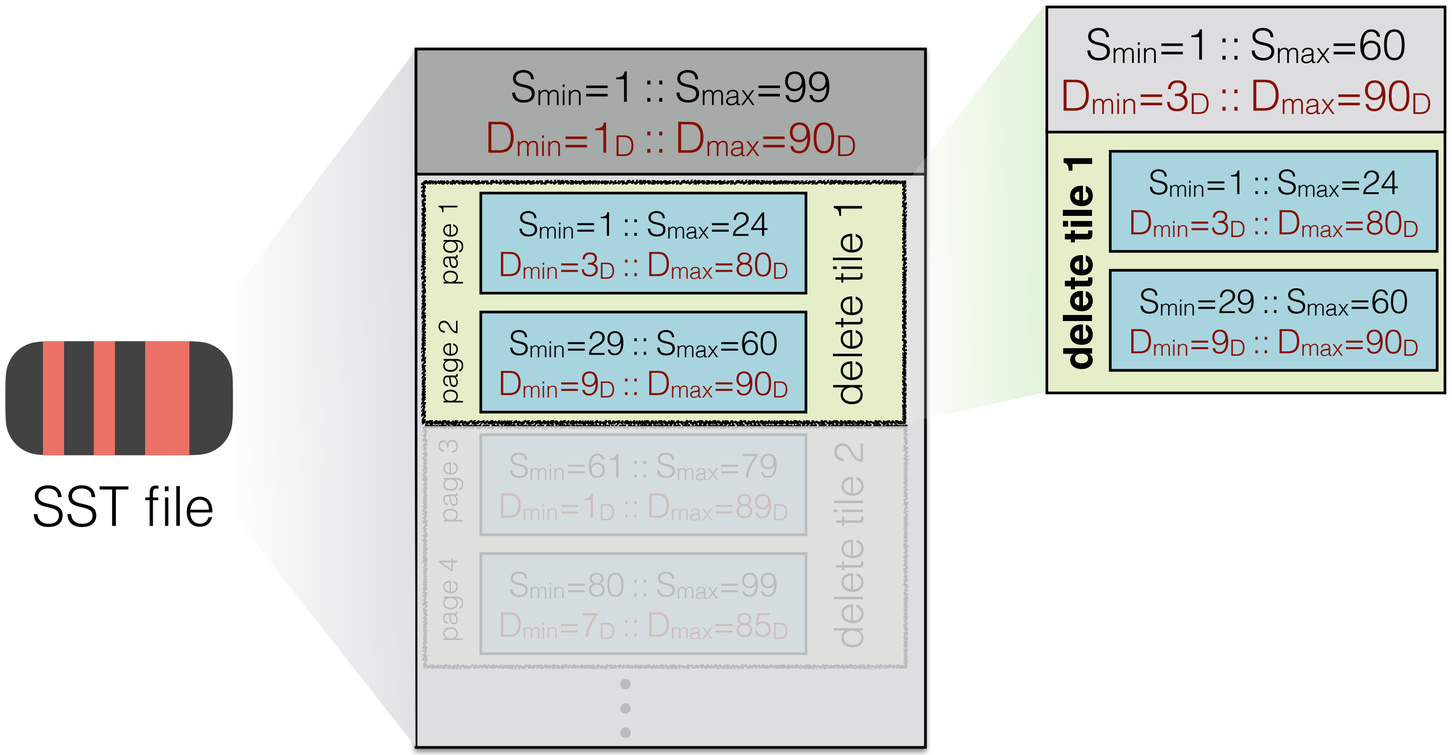
Key Weaving storage layout

delete all entries older than <= 65_D



partitioned on S

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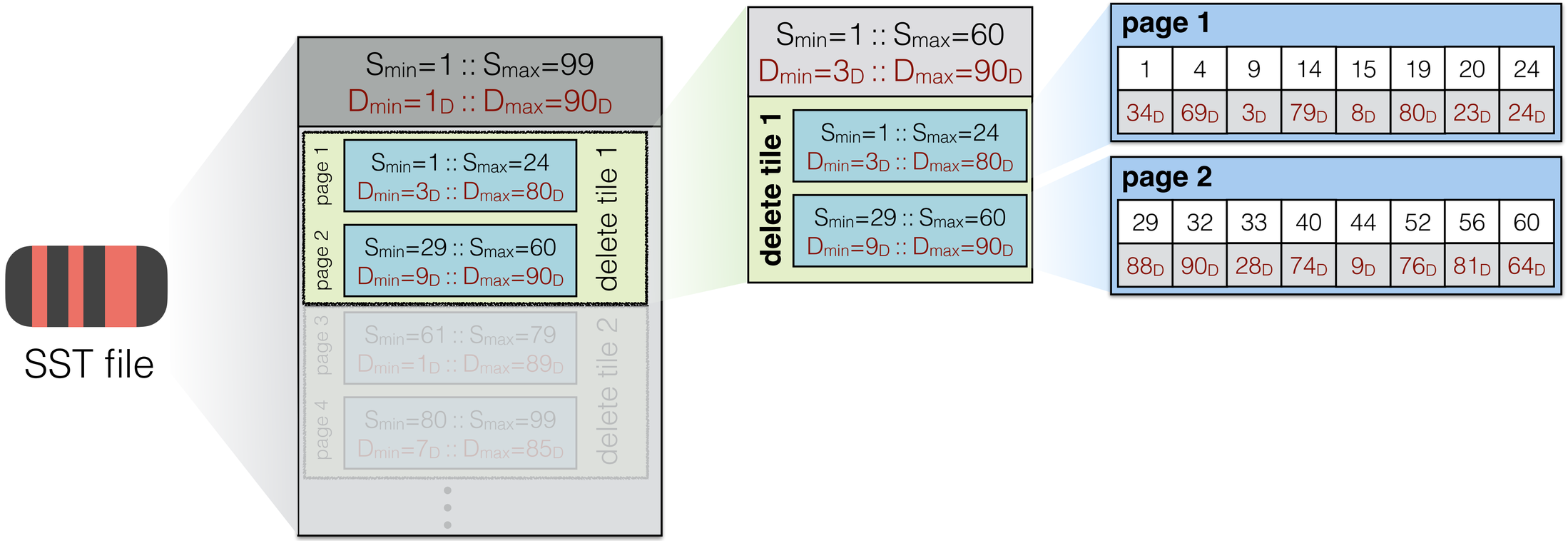
Key Weaving storage layout

delete all entries older than <= 65_D



partitioned on S

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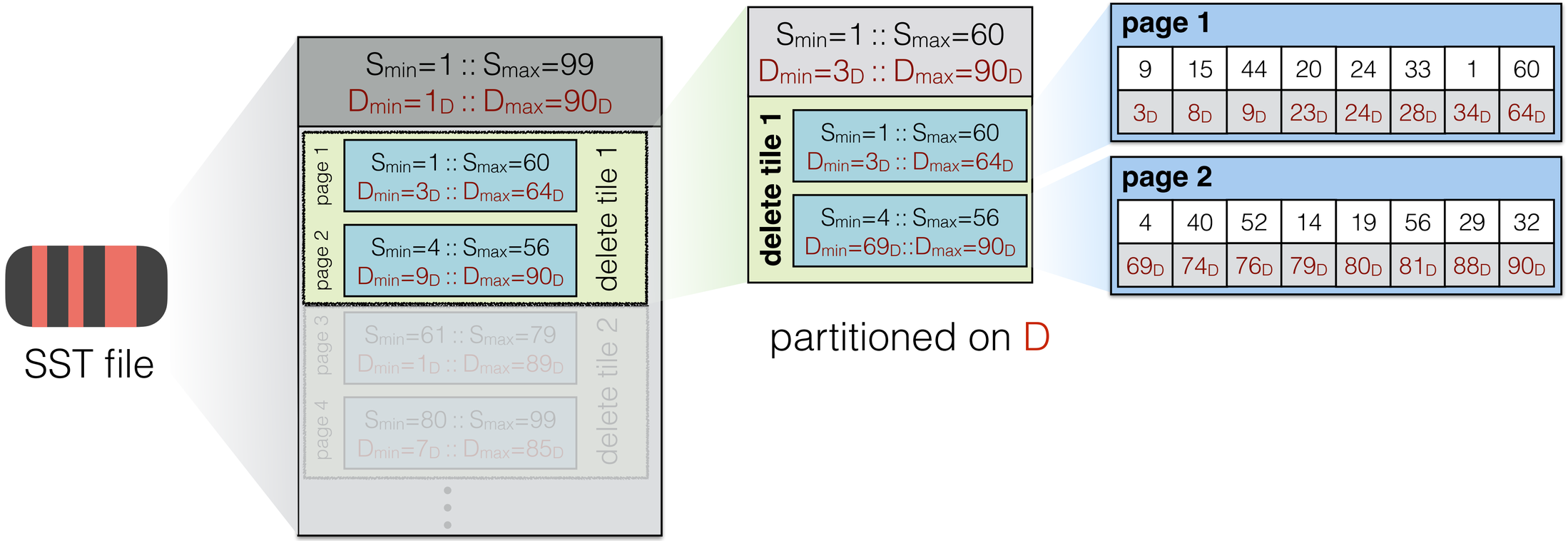
Key Weaving storage layout

delete all entries older than <= 65_D



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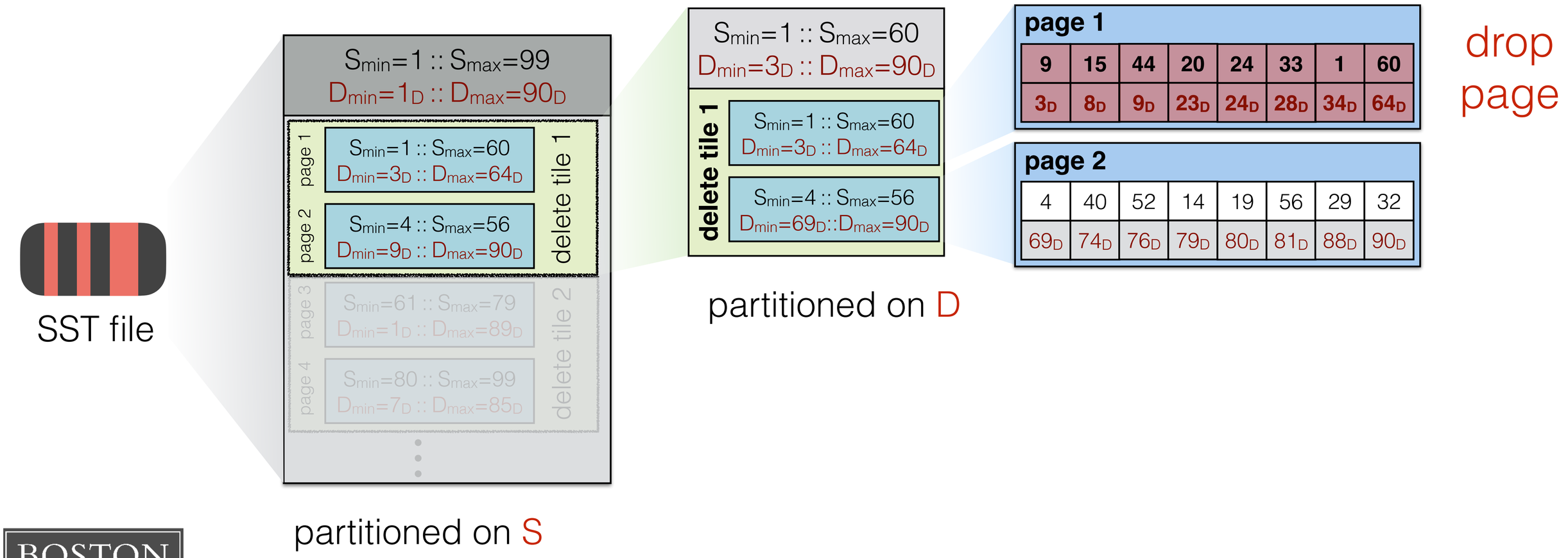
Key Weaving storage layout

delete all entries older than <= 65_D



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Key Weaving storage layout

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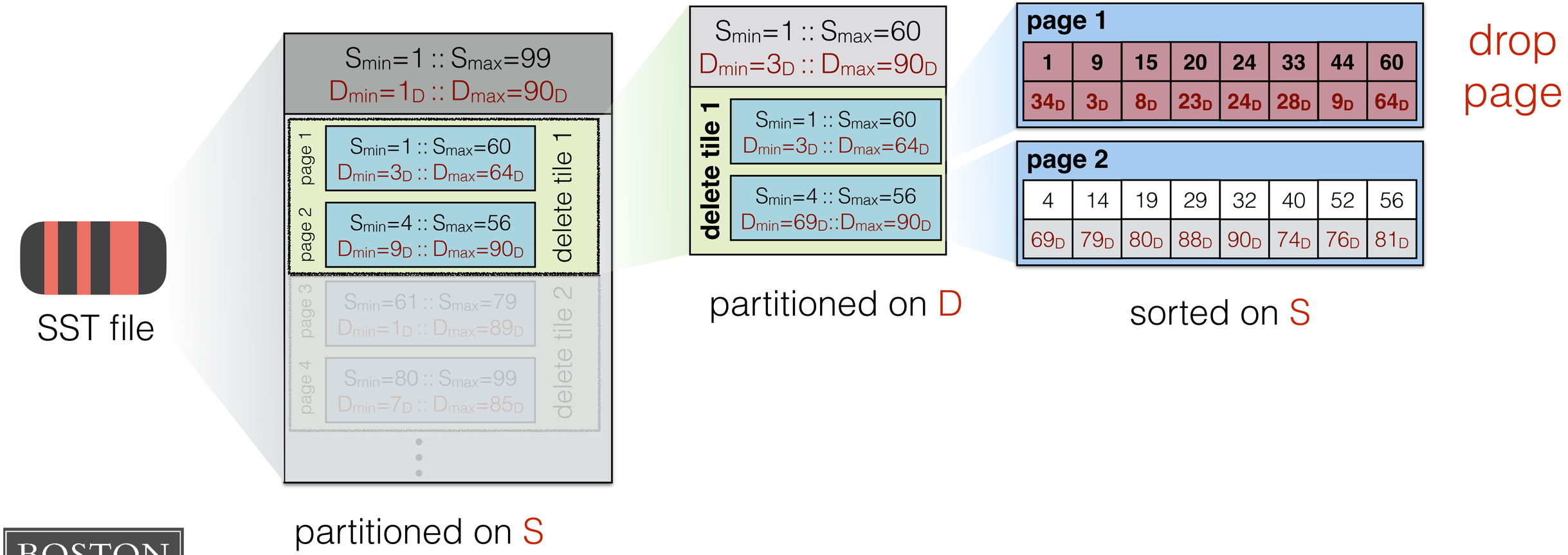

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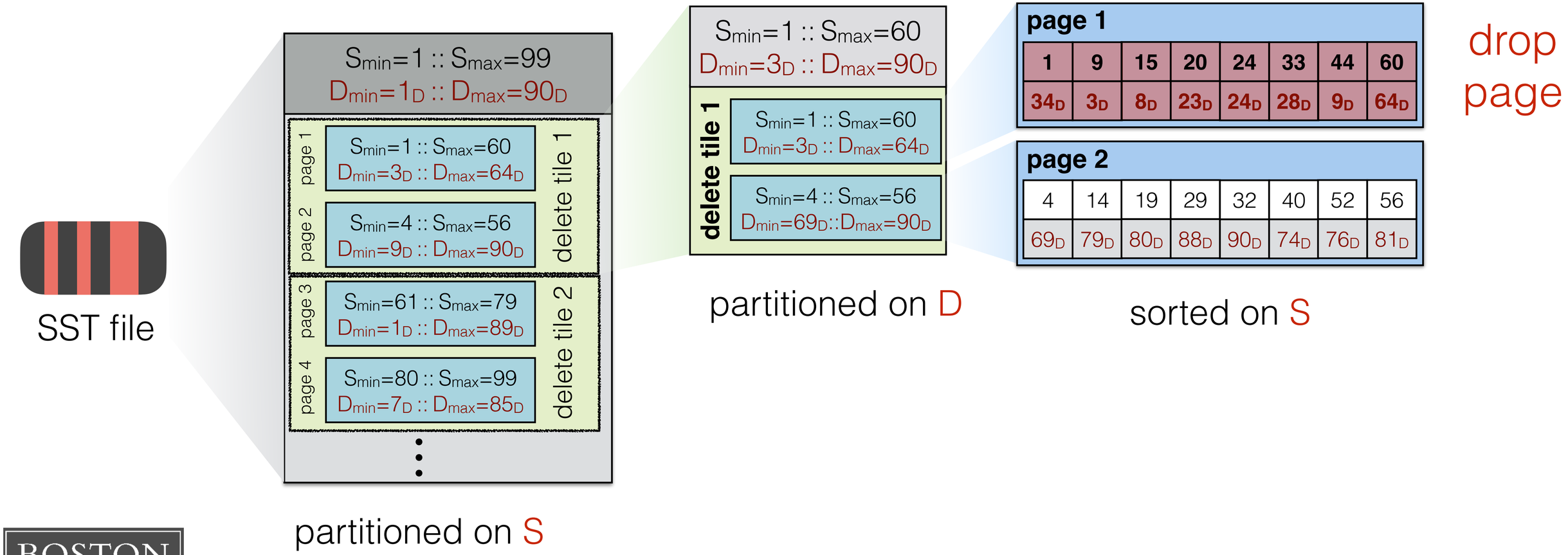
Key Weaving storage layout

delete all entries older than <= 65_D



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Key Weaving storage layout

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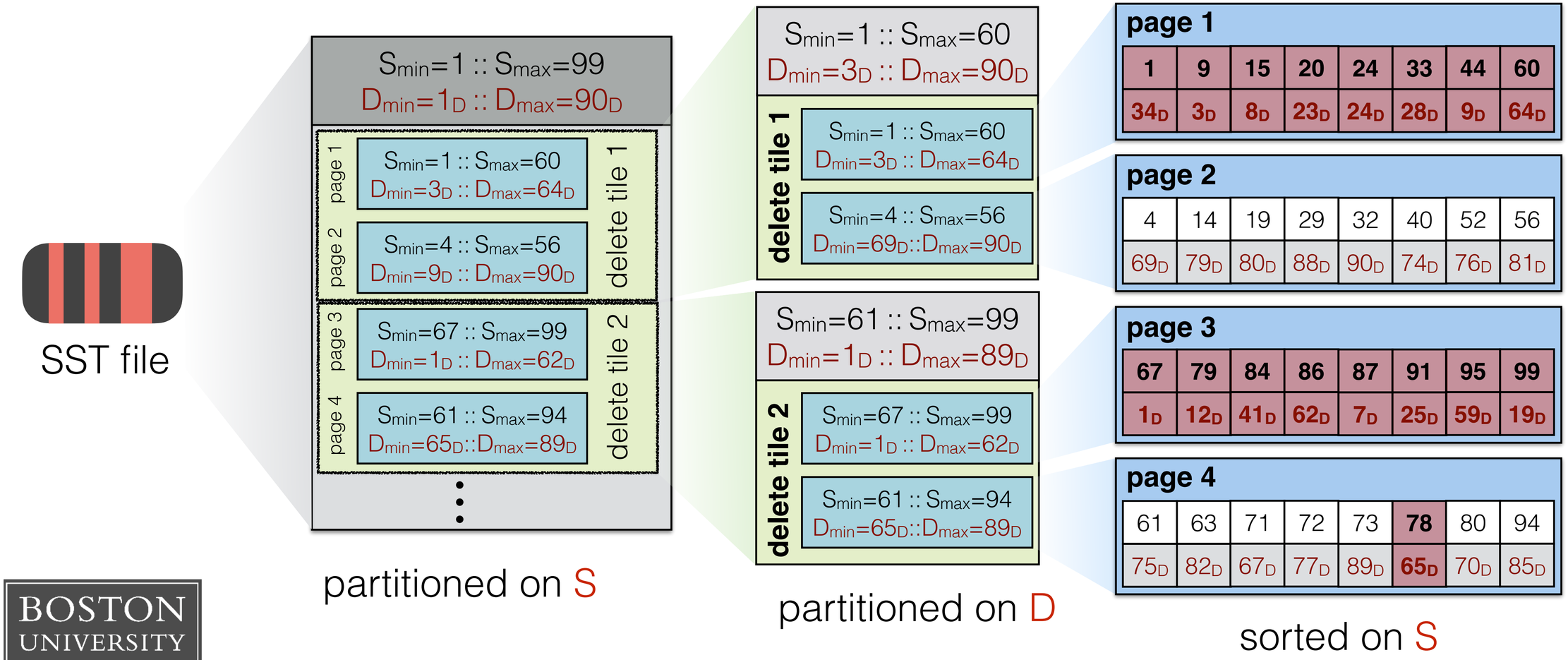

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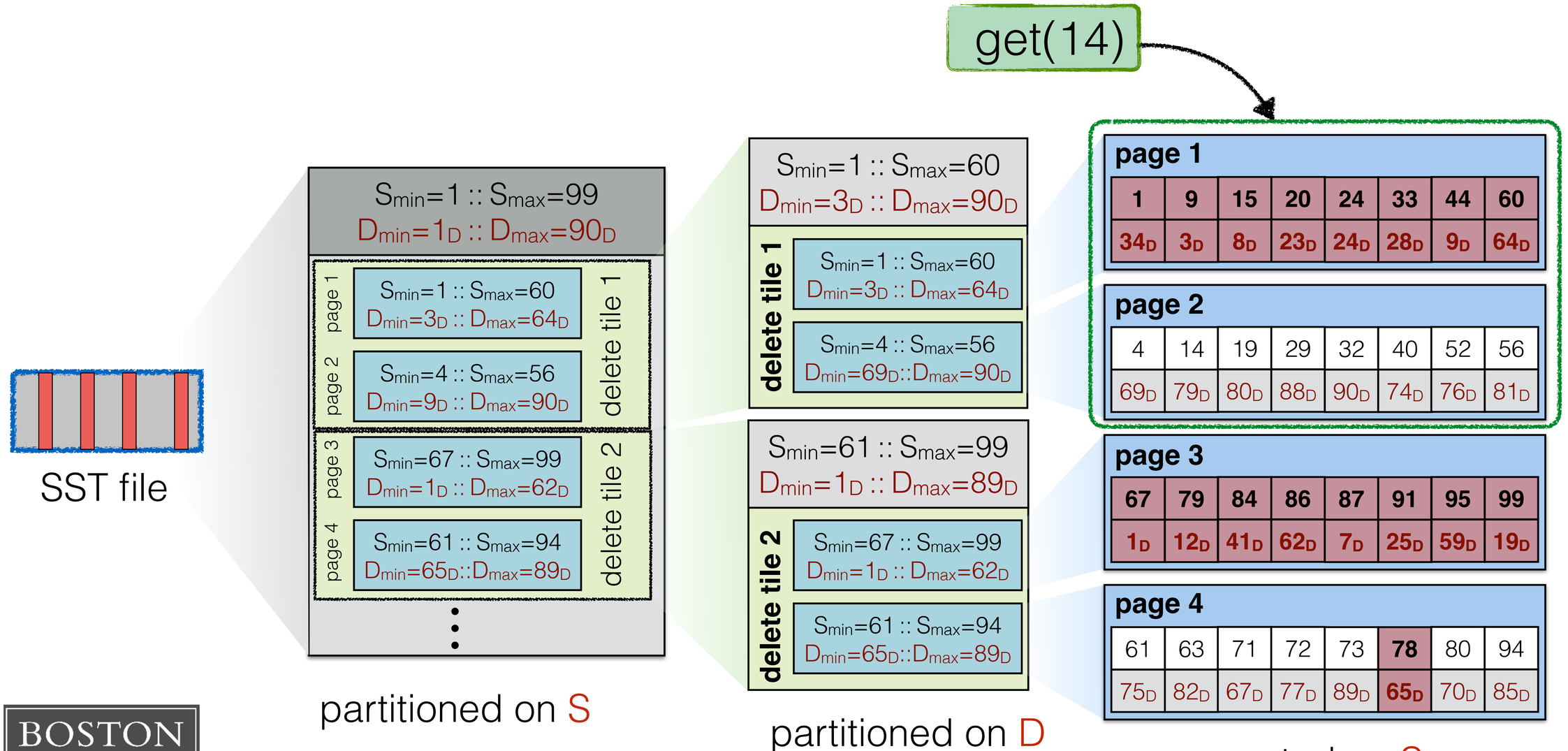
Key Weaving storage layout



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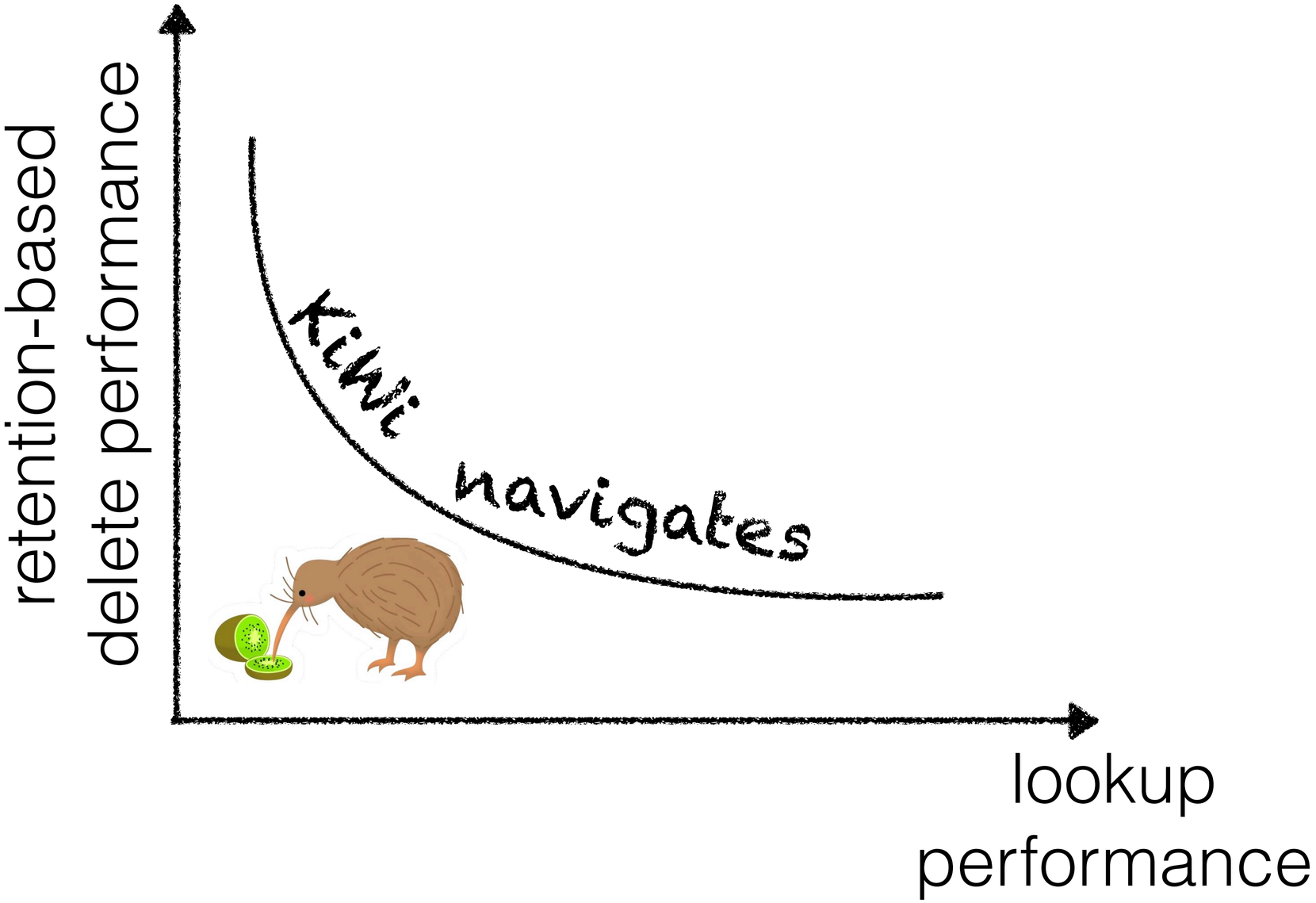

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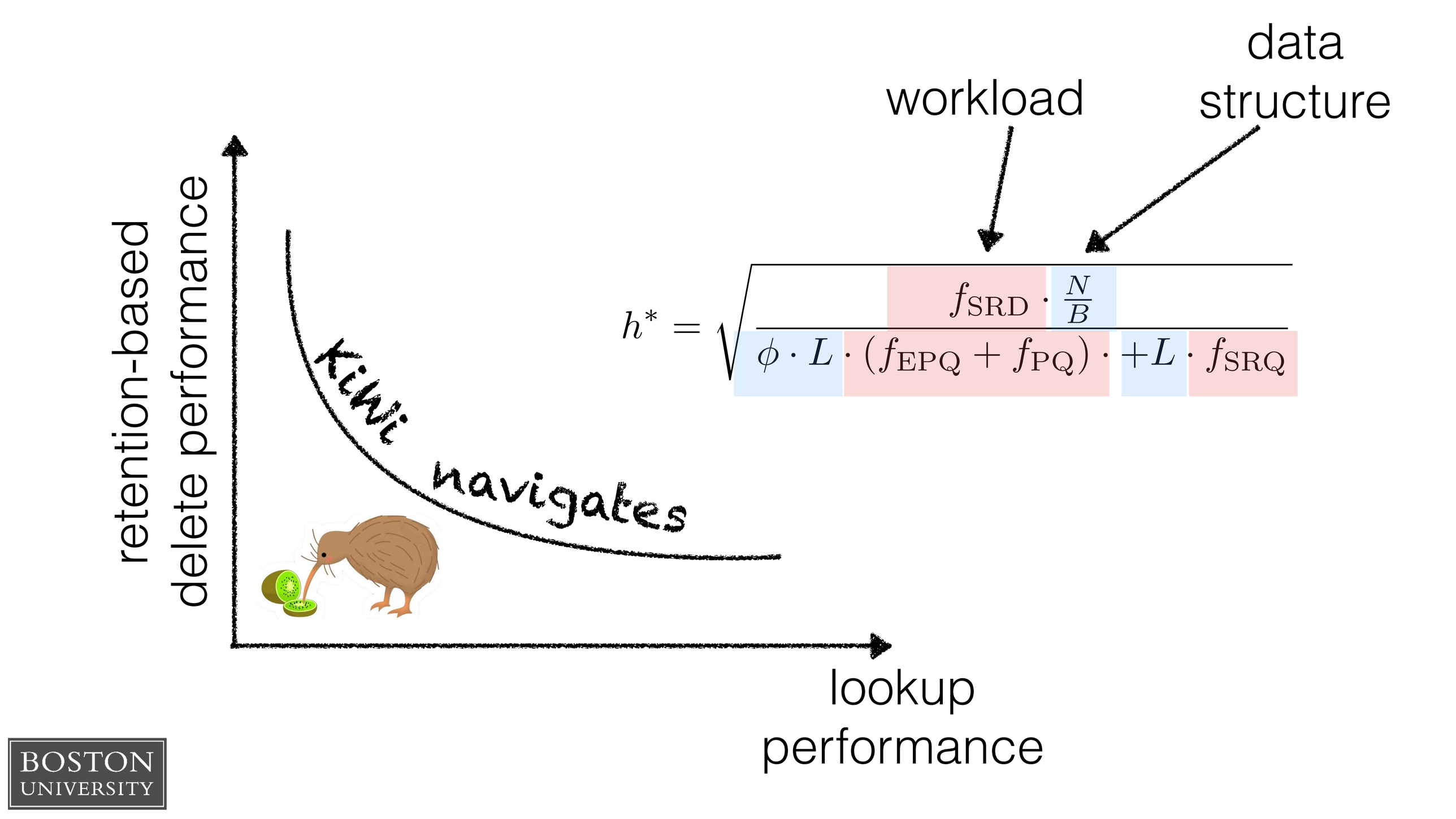

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Range Deletes

Timely Deletes from B-e trees

Persisting Updates

On-demand Deletes

Deletion in Distributed Frameworks

Deletion from File System

Deletion from Archive

Delete from View & Snapshots

Regulation Compliance as a System Feature

Secure Deletes from Hardware

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SIGMOD ’20

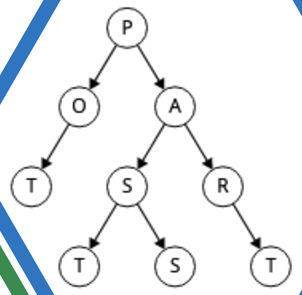
VLDB ’21

EDBT ’22

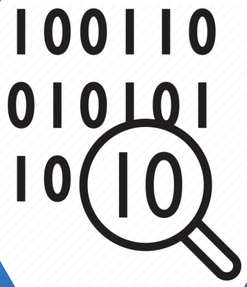
DEBull ’22

SIGMOD ’22

Timely Deletes from B-e trees



Deletion in Distributed Frameworks



Regulation Compliance as a System Feature

Deletion from File System

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https://disc.bu.edu/persistent-deletes

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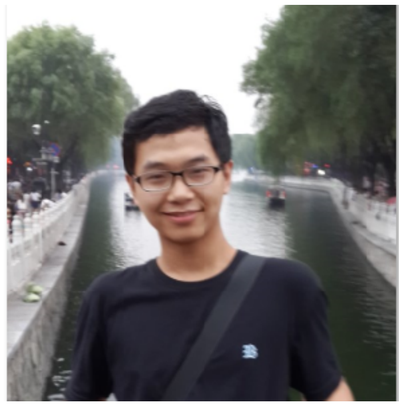
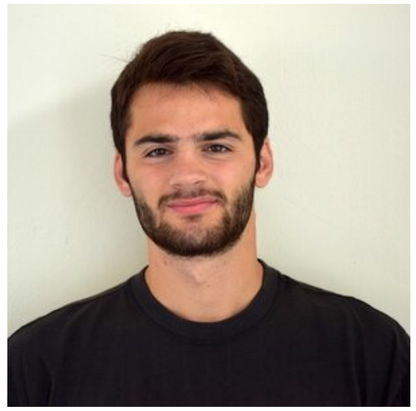
My **Amazing** Collaborators

Thank You!

https://disc.bu.edu/persistent-deletes

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NSF

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